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(54) **CARRIER HEAD AND SUBSTRATE
POLISHING APPARATUS INCLUDING THE
SAME**

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(2013.01)

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(57)

ABSTRACT

Provided are carrier heads that include a carrier body; and a
retainer ring coupled to the carrier body. A weight adjusting
groove recessed into the retainer ring by a predetermined
depth is disposed at an upper surface of the retainer ring.

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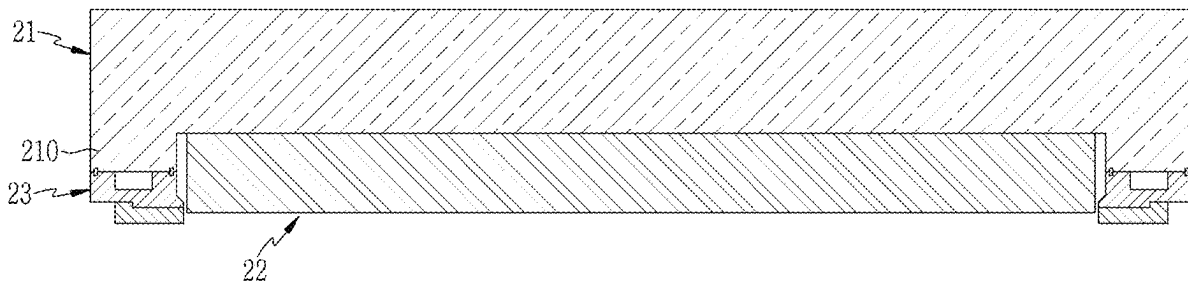


FIG. 1

1

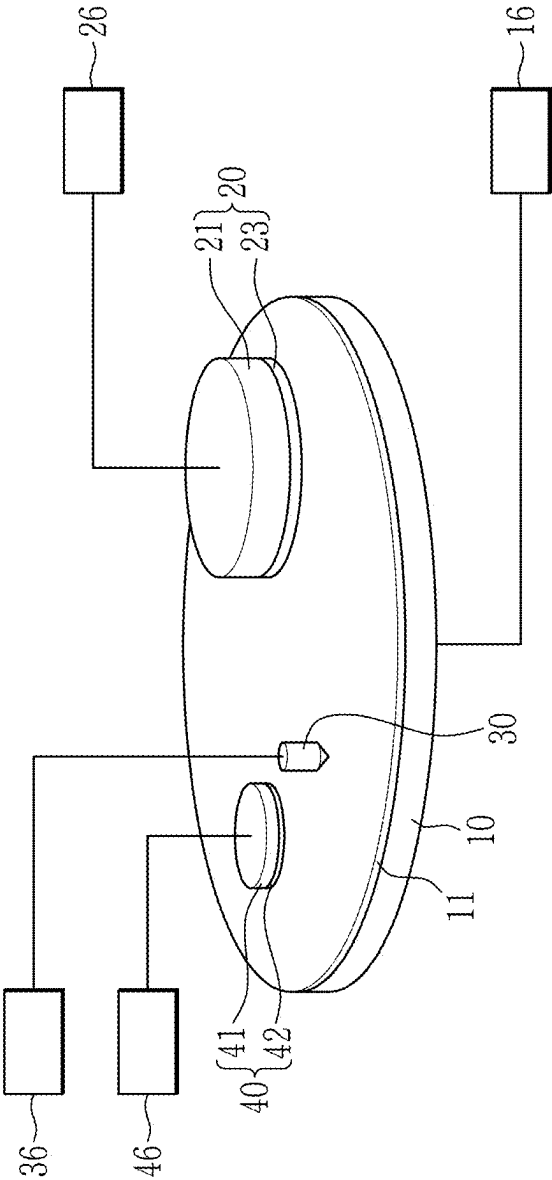


FIG. 2

20

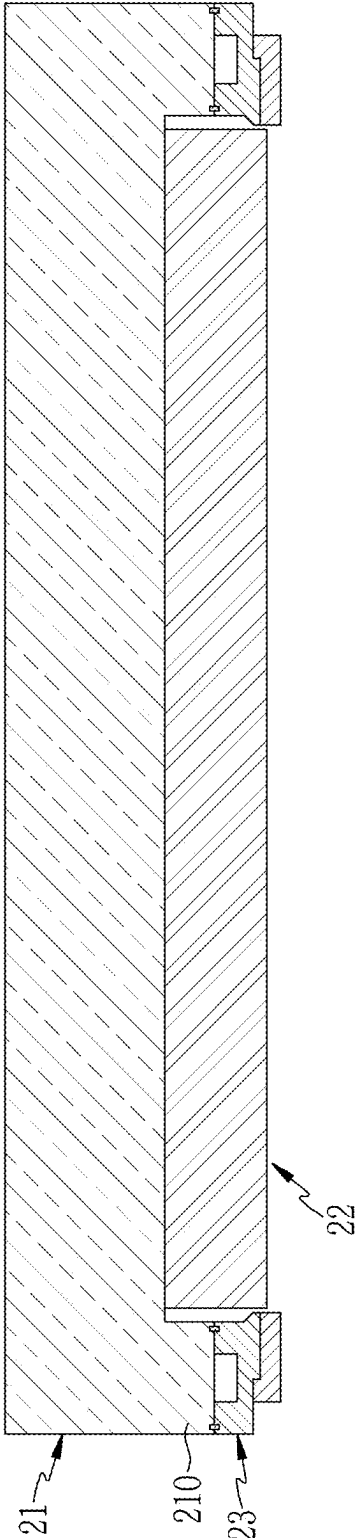


FIG. 3

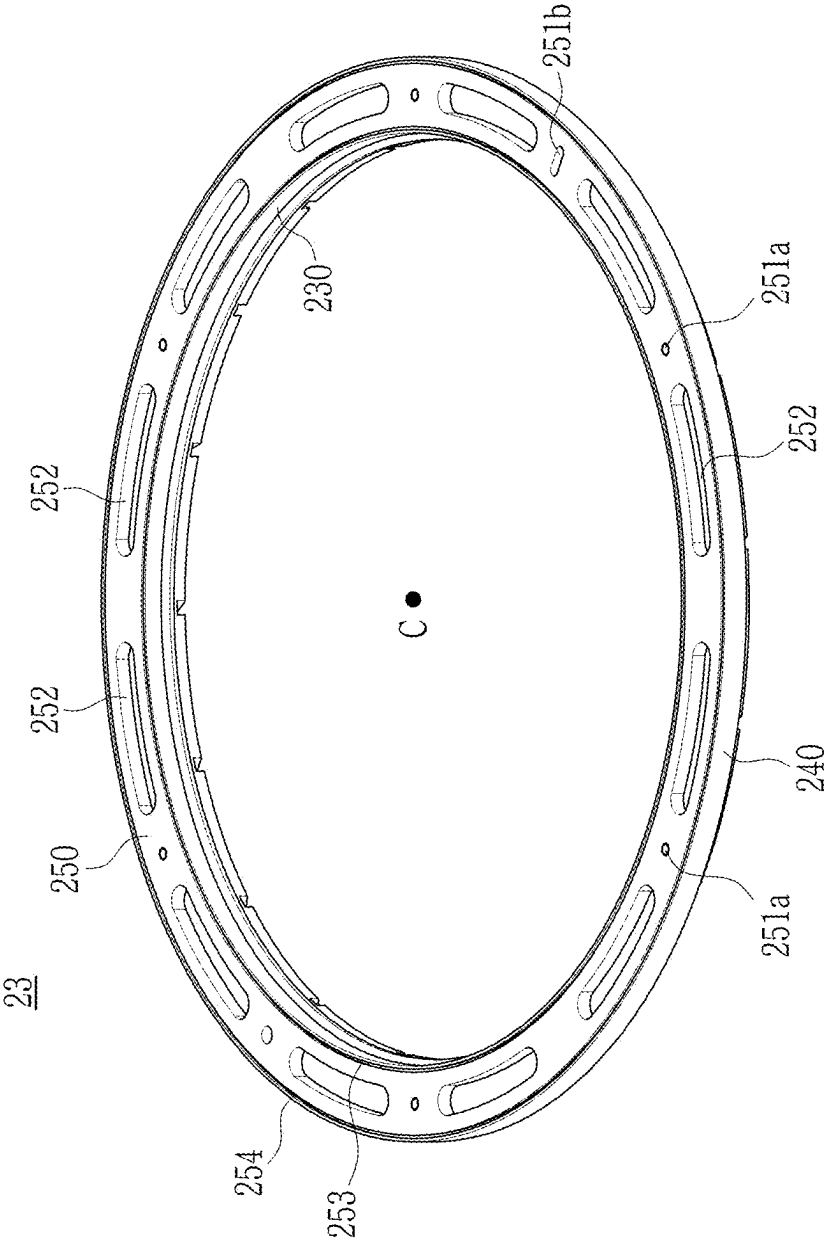


FIG. 4

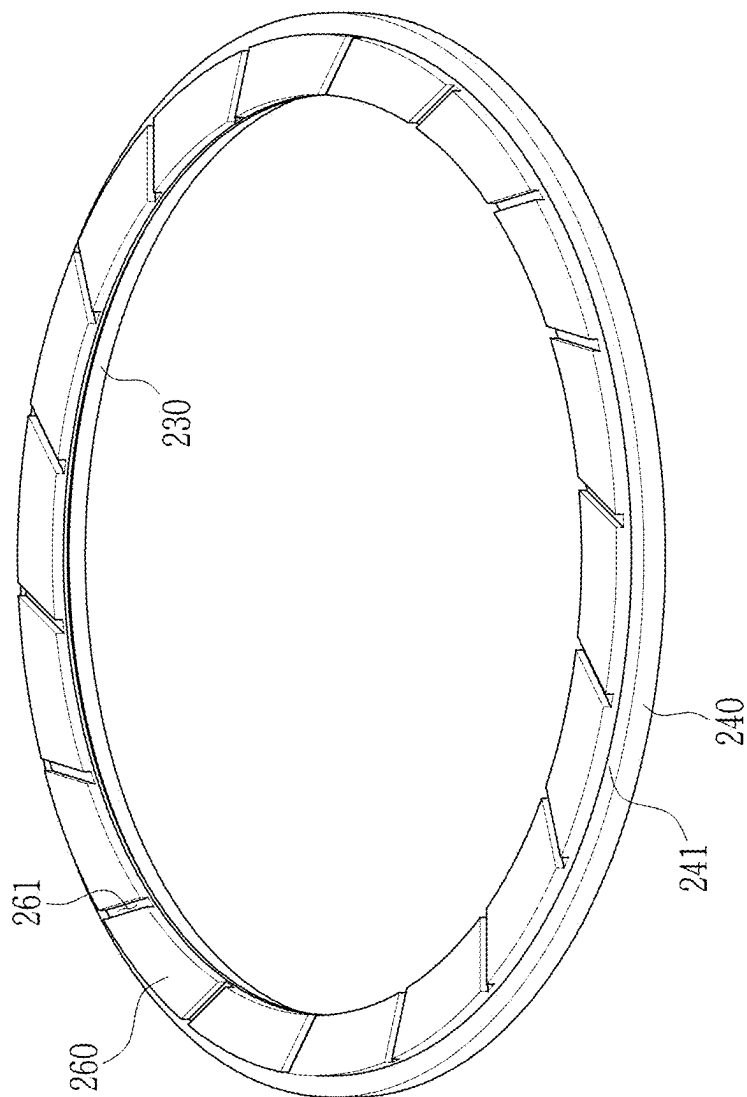
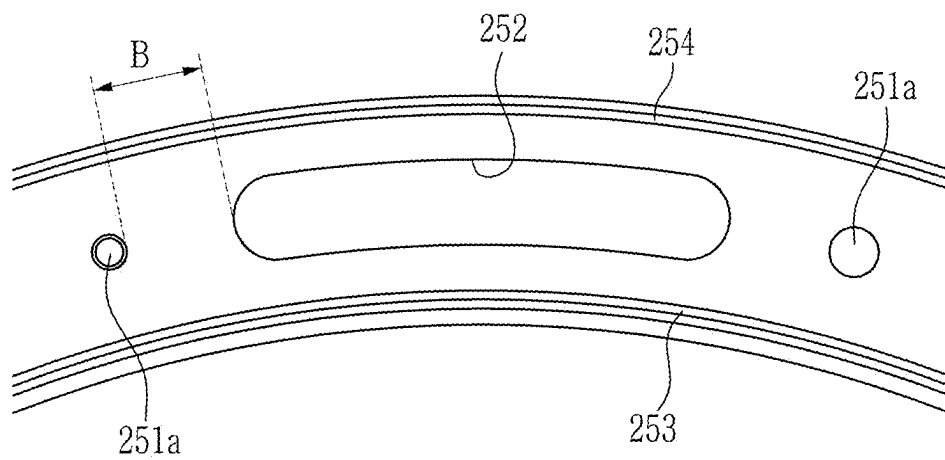


FIG. 5



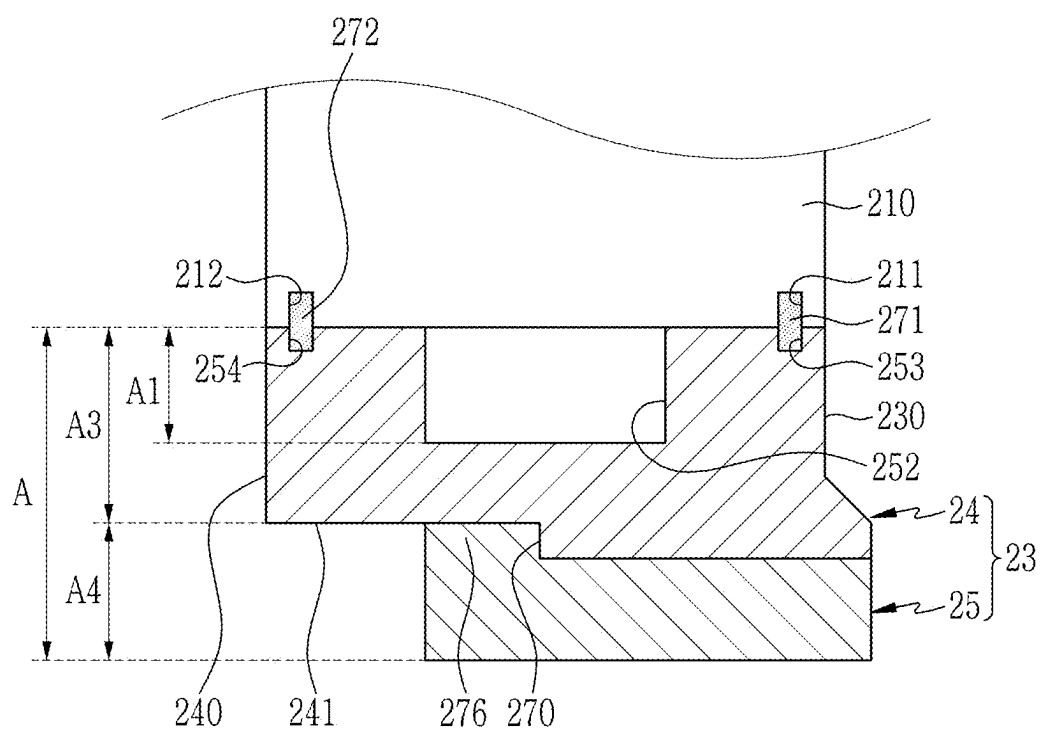


FIG. 7

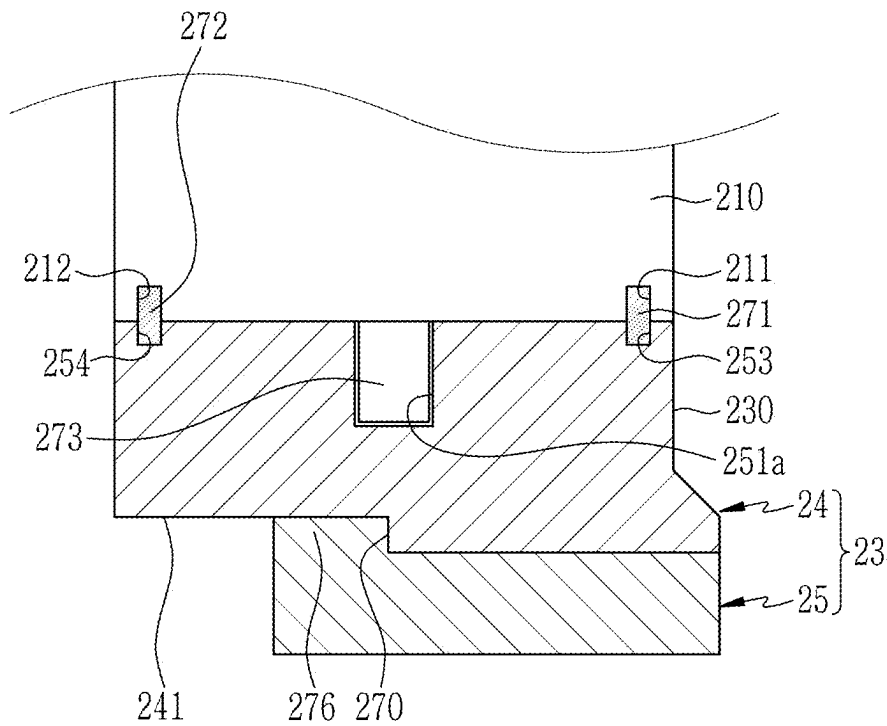


FIG. 8

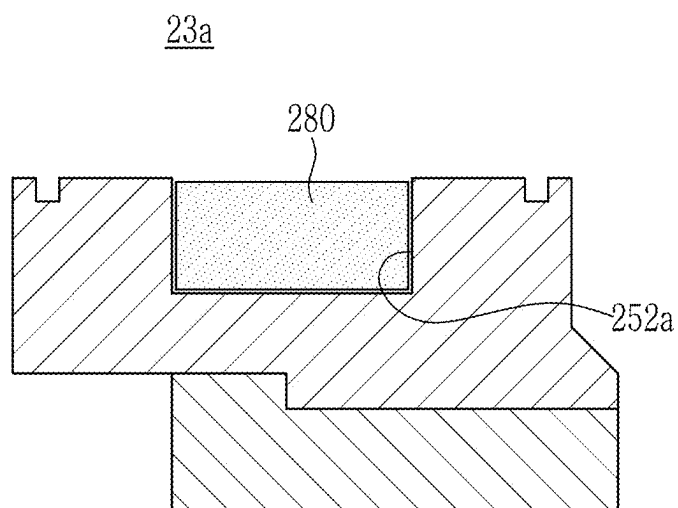


FIG. 9

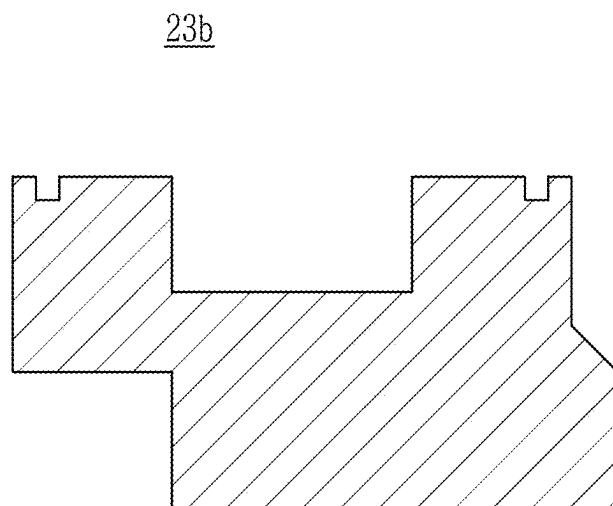


FIG. 10

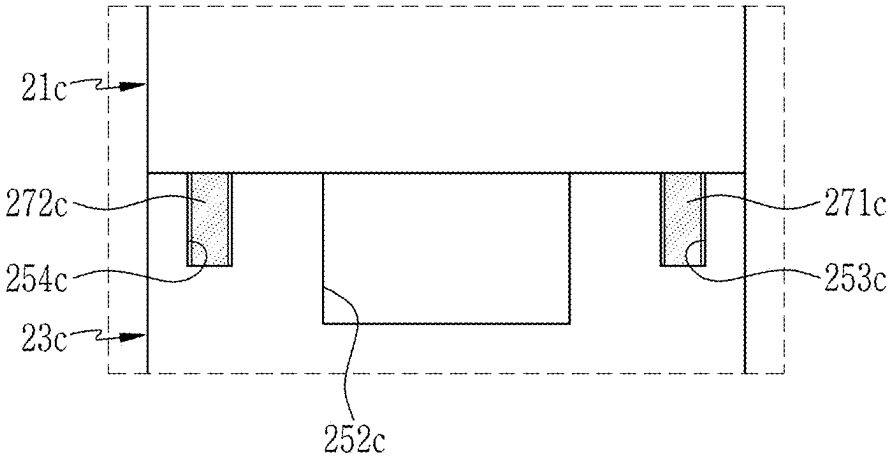


FIG. 11

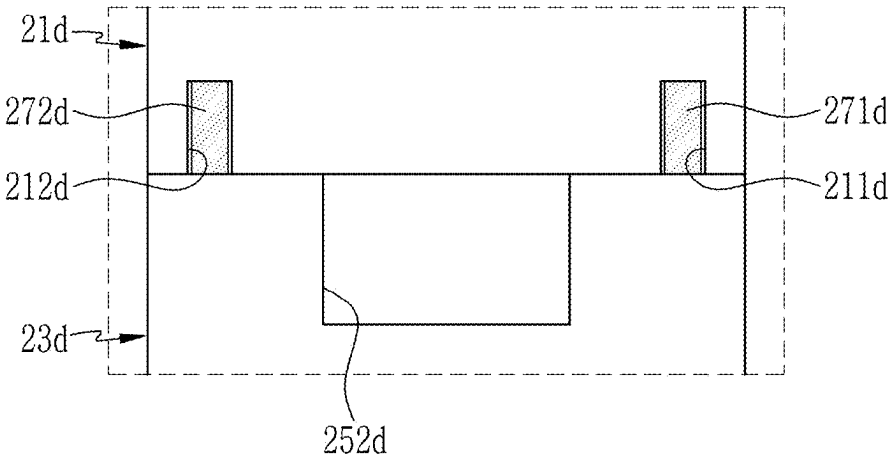


FIG. 12

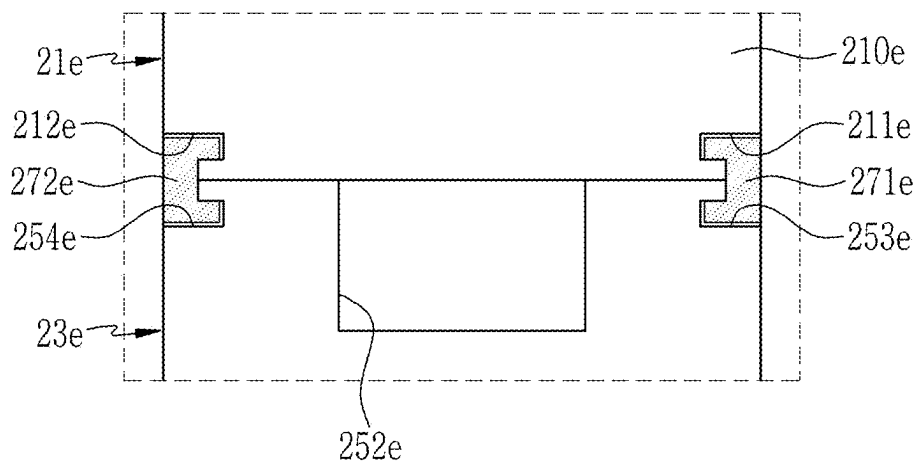


FIG. 13

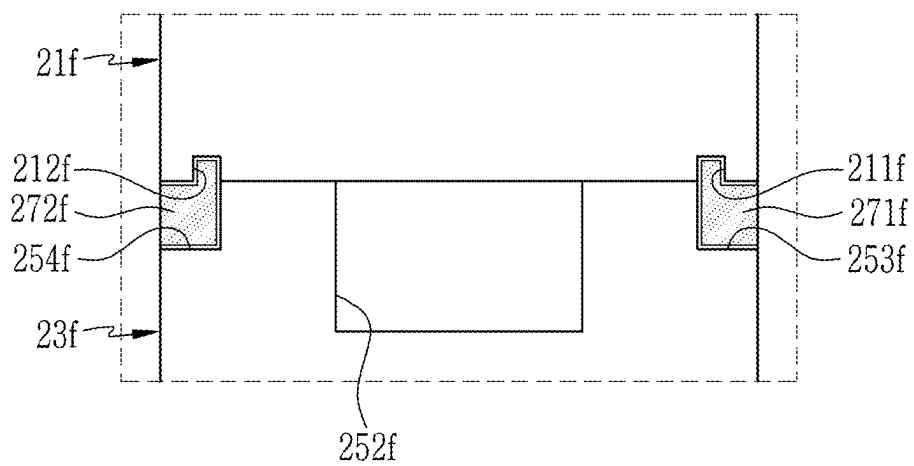


FIG. 14

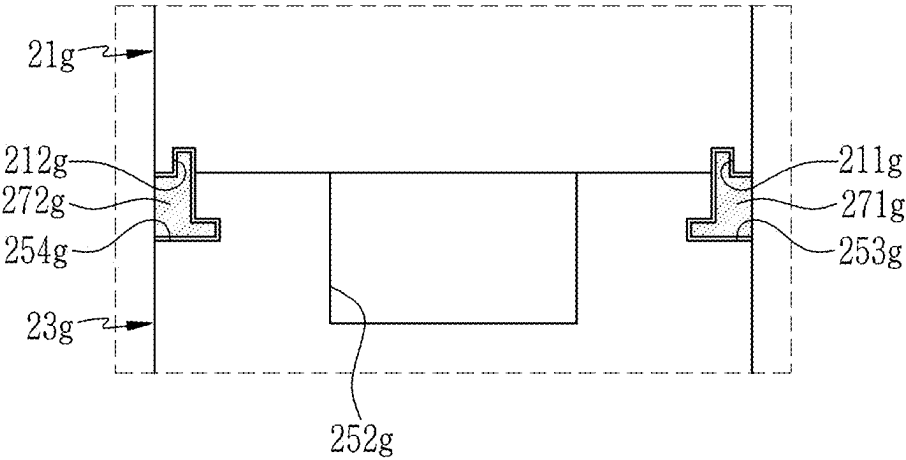


FIG. 15

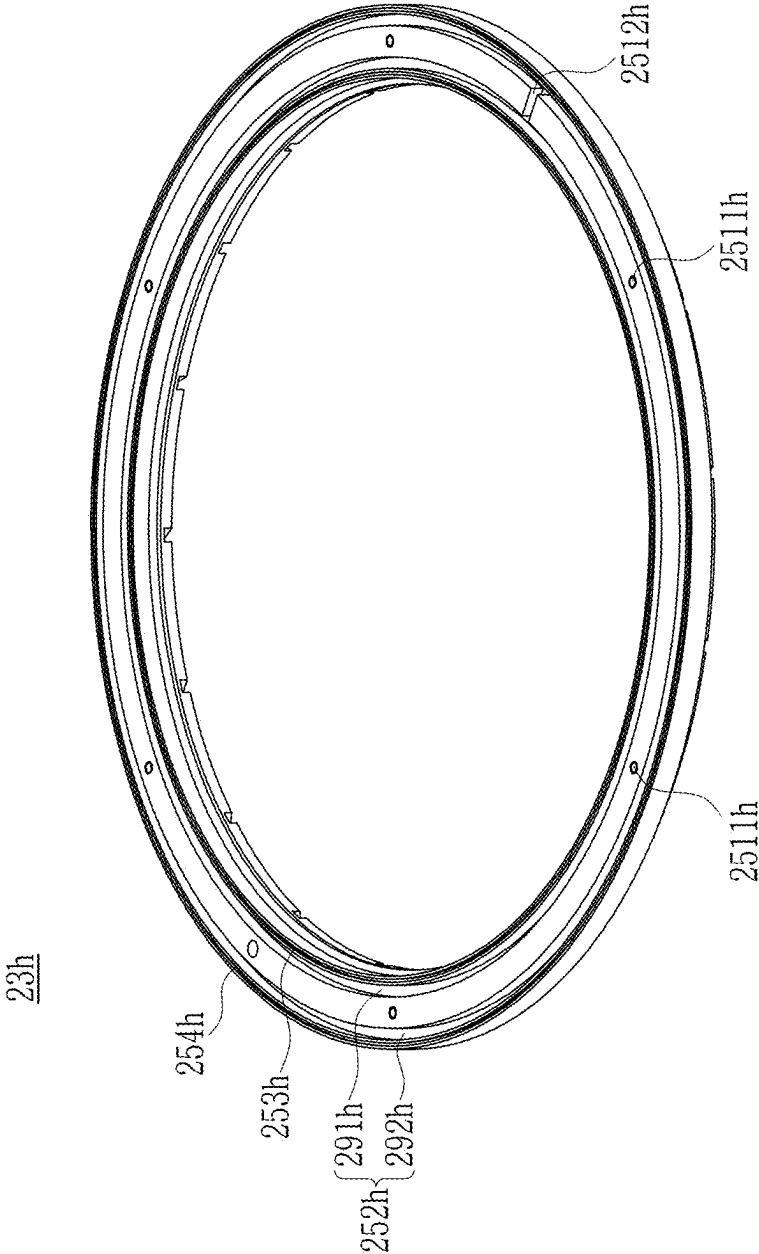


FIG. 16

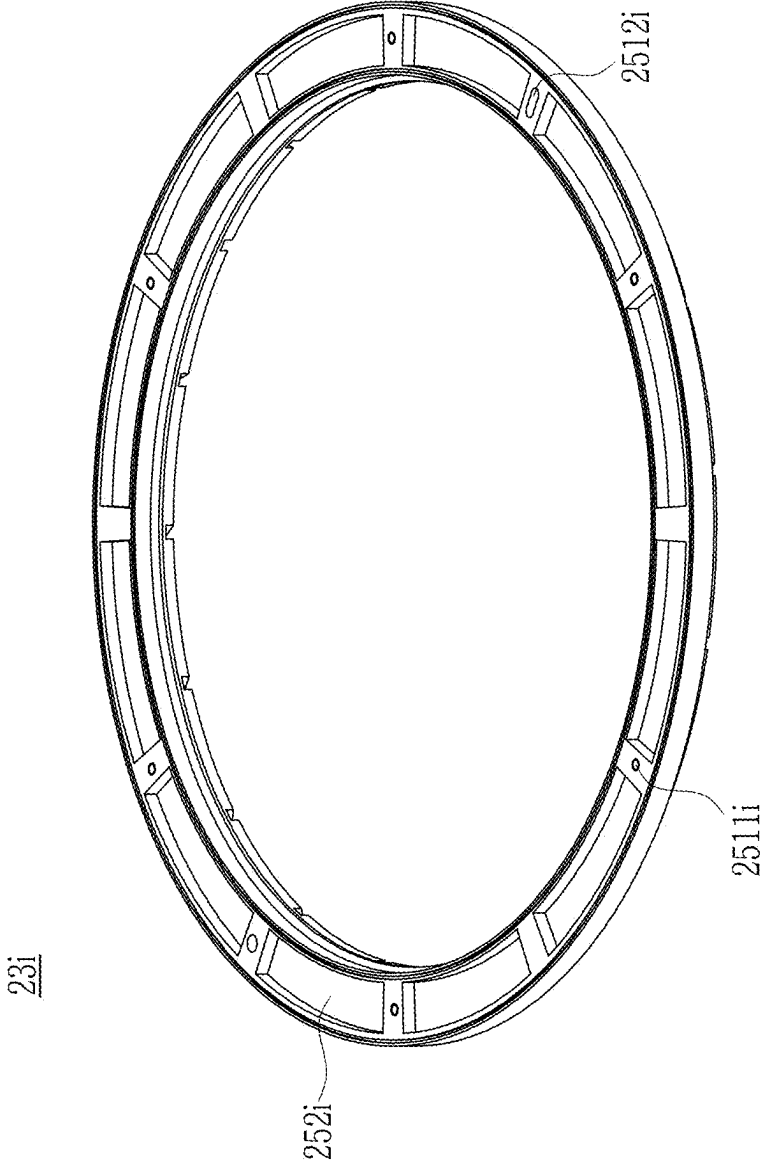


FIG. 17

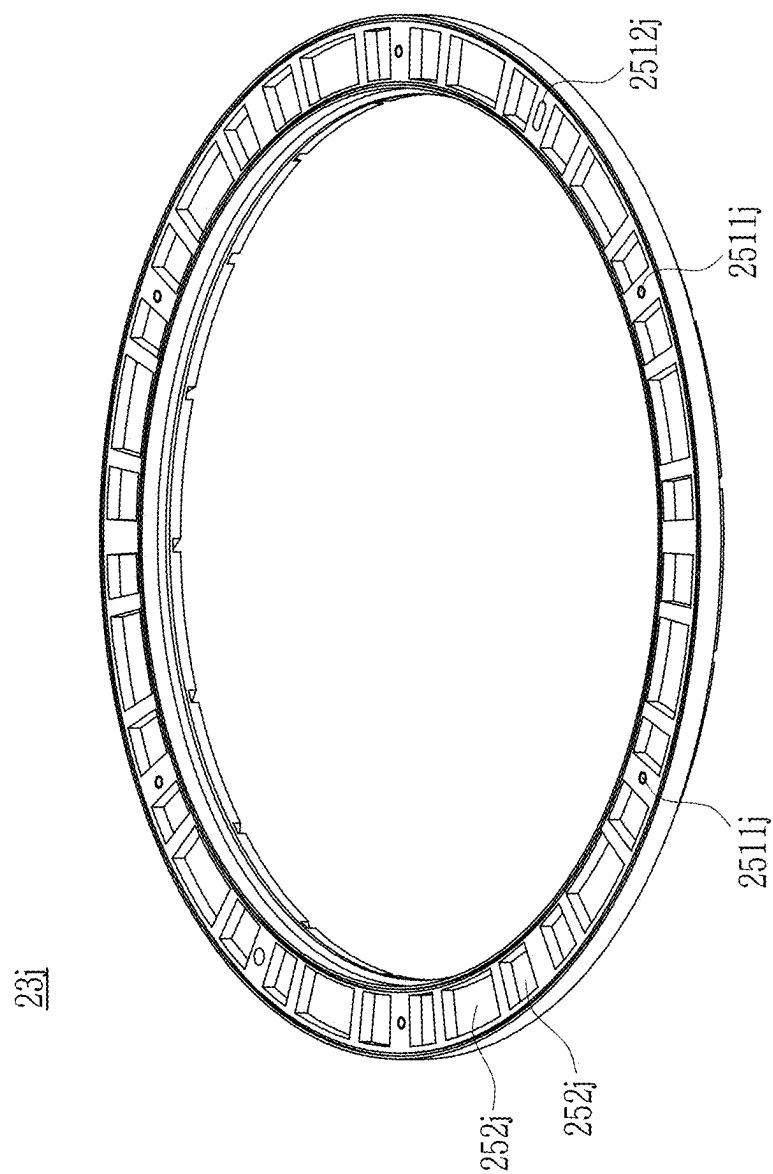


FIG. 18

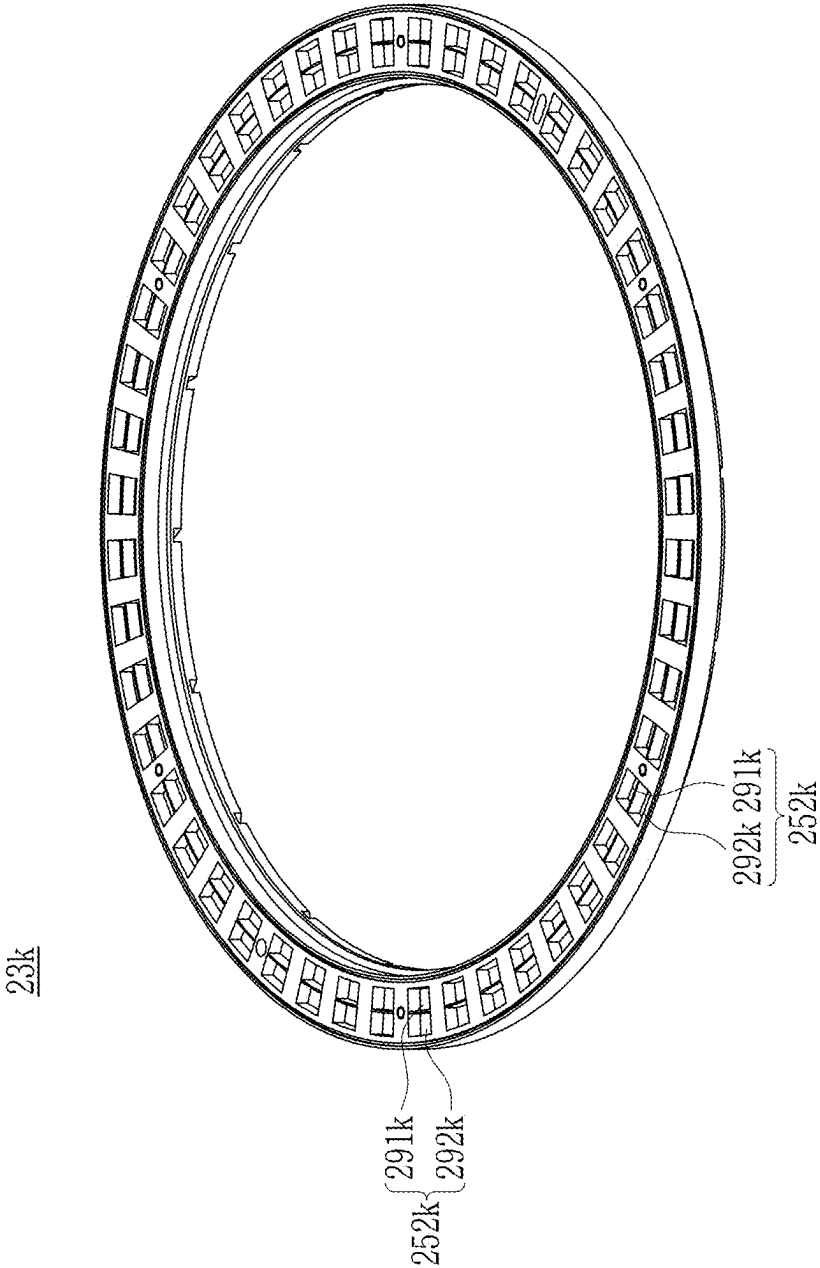
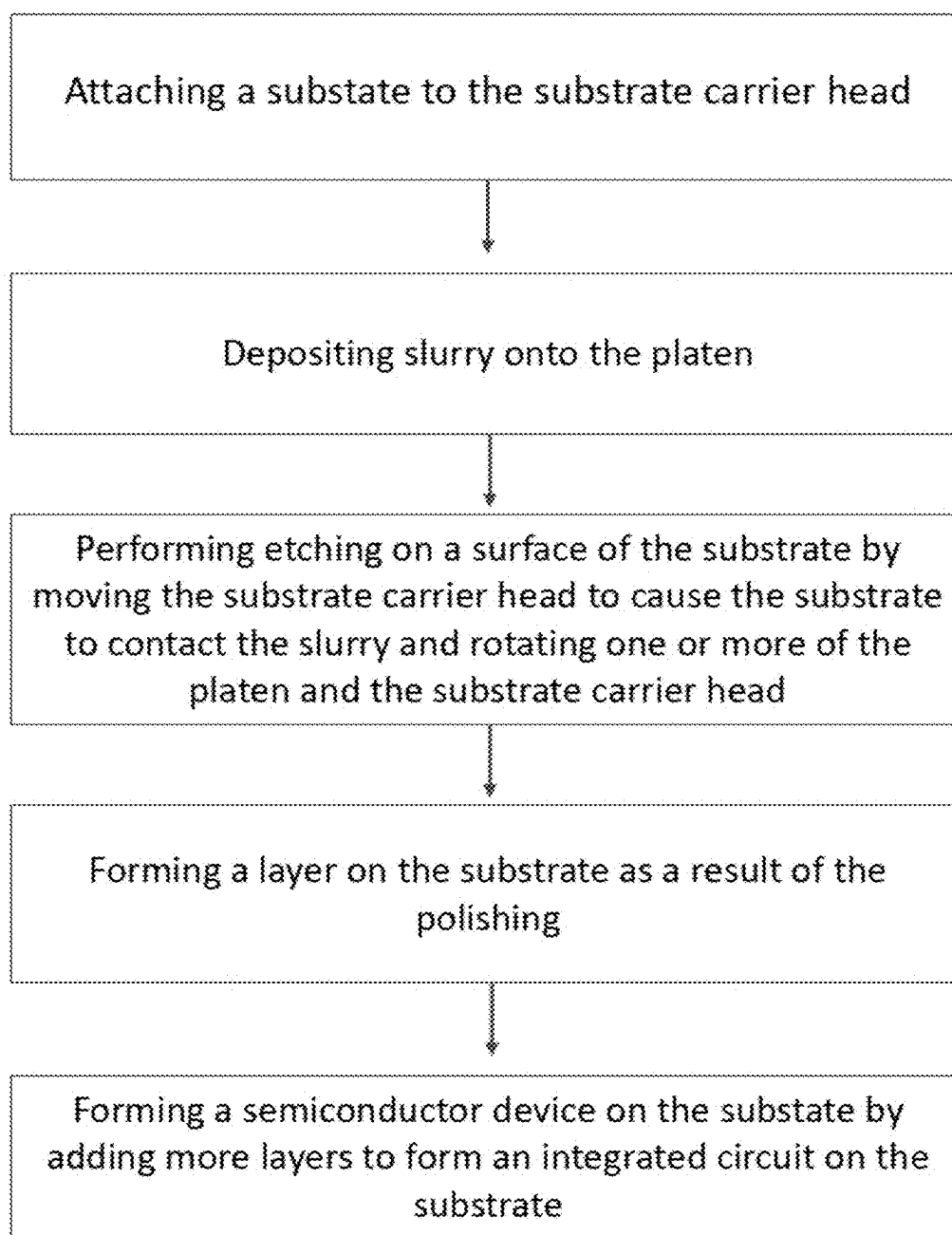


FIG. 19



CARRIER HEAD AND SUBSTRATE POLISHING APPARATUS INCLUDING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This U.S. nonprovisional application claims priority under 35 U.S.C. § 119 to and the benefit of Korean Patent Application No. 10-2024-0027112 filed at the Korean Intellectual Property Office on Feb. 26, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates to a carrier head for chemical mechanical polishing (CMP) processes, and a substrate polishing apparatus including the same.

BACKGROUND

[0003] Semiconductor device manufacturing processes include a deposition process of forming a thin film layer on a wafer and an etching process of forming a fine circuit pattern on the thin film layer. The processes are repeated until a required circuit pattern is formed on the wafer, and after the circuit pattern is formed, many curvatures are formed on a surface of the wafer. A structure of a semiconductor device has recently become multilayered according to high integration, and the number of curvatures on the surface of the wafer and a step difference (or a height difference) between the curvatures are increasing. However, because non-planarization of the surface of the wafer causes problems such as defocus or the like in a photolithography process, periodic polishing is required to planarize the surface of the wafer.

[0004] There are various surface planarization techniques to planarize the surface of wafers. Among these, a chemical mechanical polishing (CMP) technique is widely used because a wide surface as well as a narrow surface may be planarized with excellent flatness by using the CMP technique. A chemical mechanical polishing device includes a carrier head that applies pressure to a polishing pad while supporting the wafer.

SUMMARY

[0005] Example embodiments of the present invention provide a substrate carrier head capable of effectively polishing a substrate, which may be a package substrate, as part of a CMP process, and a substrate polishing apparatus including the same.

[0006] A substrate carrier head according to example embodiments includes: a carrier body, and a retainer ring coupled to the carrier body, wherein the retainer ring includes a weight adjusting groove recessed into the retainer ring by a predetermined depth. The retainer ring has an upper surface facing the carrier body. The weight adjusting groove is disposed at the upper surface of the retainer ring.

[0007] A substrate carrier head according to example embodiments includes: a carrier body; a retainer ring coupled to the carrier body; a fastening member that protrudes from the carrier body toward the retainer ring which fastening member is configured to fit into a fastening groove of the retainer ring; and a seal ring disposed between the retainer ring and the carrier body. The retainer ring has an upper surface facing the carrier body. The fastening groove

recessed into the retainer ring by a predetermined depth, and a weight adjusting groove recessed into the retainer ring by a predetermined depth are disposed at the upper surface of the retainer ring.

[0008] A substrate polishing apparatus according to example embodiments include: a platen; and a substrate carrier head disposed adjacent to the platen. The substrate carrier head includes a carrier body, a retainer ring coupled to the carrier body, a fastening member that protrudes from the carrier body toward the retainer ring, which fastening member is configured to fit into a fastening groove of the retainer ring, an inner seal ring disposed between the retainer ring and the carrier body at a region between a weight adjusting groove and an upper end portion of an inner surface of the retainer ring, and an outer seal ring disposed between the retainer ring and the carrier body at a region between the weight adjusting groove and an upper end portion of an outer surface of the retainer ring. The retainer ring has an upper surface facing the carrier body. The fastening groove recessed into the retainer ring by a predetermined depth r and a weight adjusting groove recessed into the retainer ring by a predetermined depth, are disposed at the upper surface of the retainer ring. According to embodiments, the fastening member is inserted into the fastening groove.

[0009] According to the embodiments, a carrier head capable of effectively polishing a substrate and a substrate polishing apparatus including the same may be provided.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a view showing a substrate polishing apparatus according to example embodiments.

[0011] FIG. 2 is a longitudinal cross-sectional view of a carrier head of FIG. 1, which carrier head includes a retainer ring.

[0012] FIG. 3 is an upper perspective view of a retainer ring of FIG. 2.

[0013] FIG. 4 is a lower perspective view of the retainer ring of FIG. 2.

[0014] FIG. 5 is a partial plan view of the retainer ring of FIG. 3.

[0015] FIG. 6 is a longitudinal cross-sectional view of a lower outer side of the carrier head passing through a region in which a weight adjusting groove is disposed at the retainer ring.

[0016] FIG. 7 is a longitudinal cross-sectional view of a lower outer side of the carrier head passing through a region in which a fastening groove is disposed at the retainer ring.

[0017] FIG. 8 is a cross-sectional view of a retainer ring according to another embodiment.

[0018] FIG. 9 is a cross-sectional view of a retainer ring according to another embodiment.

[0019] FIG. 10 is a view showing a disposition structure of a seal ring according to another embodiment.

[0020] FIG. 11 is a view showing a disposition structure of a seal ring according to another embodiment.

[0021] FIG. 12 is a view showing a disposition structure of a seal ring according to another embodiment.

[0022] FIG. 13 is a view showing a disposition structure of a seal ring according to another embodiment.

[0023] FIG. 14 is a view showing a disposition structure of a seal ring according to another embodiment.

[0024] FIG. 15 is an upper perspective view of a retainer ring according to another embodiment.

[0025] FIG. 16 is an upper perspective view of a retainer ring according to another embodiment.

[0026] FIG. 17 is an upper perspective view of a retainer ring according to another embodiment.

[0027] FIG. 18 is an upper perspective view of a retainer ring according to another embodiment.

[0028] FIG. 19 is a flow chart depicting an example manufacturing method in accordance with example embodiments.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0029] Embodiments of the present disclosure will be described more fully hereinafter with reference to the accompanying drawings so that those skilled in the art easily implement the embodiments. The present disclosure may be modified in various different ways, all without departing from the spirit or scope of the present disclosure. Items described in the singular herein may be provided in plural. Thus, the description of a single item that is provided in plural should be understood to be applicable to the remaining plurality of items unless context indicates otherwise.

[0030] To clearly describe the present disclosure, parts or portions of certain features are omitted, and identical or similar constituent elements throughout the specification are denoted by the same reference numerals.

[0031] Further, in the drawings, the size and thickness of each element are illustrated for ease of description, and the present disclosure is not limited to those illustrated in the drawings. In the drawings, the depictions of thicknesses of layers, films, panels, regions, areas, etc., are exaggerated for clarity, and are not intended to be limiting in any way.

[0032] It will be understood that when an element such as a layer, film, region, area, or substrate is referred to as being “on” or “above” another element, it may be directly on the other element, or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” another element, there are no intervening elements present. Further, in the specification, the word “on” or “above” means disposed on or below the object portion, and does not necessarily mean disposed on the upper side of the object portion based on a gravitational direction.

[0033] In addition, unless explicitly described to the contrary, the word “comprise” and variations such as “comprises” or “comprising” will be understood to imply the inclusion of stated elements but not the exclusion of any other elements.

[0034] Further, throughout the specification, the phrase “in a plan view” or “on a plane” means viewing a target portion from the top, and the phrase “in a cross-sectional view” or “on a cross-section” means viewing a cross-section formed by vertically cutting a target portion from the side.

[0035] Terms such as “same,” or “equal,” as used herein when referring to orientation, layout, location, shapes, sizes, compositions, amounts, or other measures do not necessarily mean an exactly identical orientation, layout, location, shape, size, composition, amount, or other measure, but are intended to encompass nearly identical orientation, layout, location, shapes, sizes, compositions, amounts, or other measures within typical variations that may occur resulting from conventional manufacturing processes.

[0036] Spatially relative terms, such as “below,” “lower,” “above,” “upper,” “top,” “bottom,” “inner,” “outer,” and the like, may be used herein for ease of description to describe

positional relationships, such as illustrated in the figures, for example. It will be understood that the spatially relative terms encompass different orientations of the device in addition to the orientation depicted in the figures. As used herein, the term “inner” relates to an element e.g., at a position that is closer to an inner circumference of the retainer ring, than a corresponding “outer” element. For example, devices in accordance with present embodiments may have an “inner” weight adjusting groove closer to an inner circumference or region of the retainer ring, and an “outer” weight adjusting groove closer to an outer circumference or region of the retainer ring. As used herein, the term “upper surface” relating to the retainer ring, refers to the surface of the retaining ring facing the carrier body, regardless of the orientation of the carrier head.

[0037] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention.

[0038] FIG. 1 is a view showing a substrate polishing apparatus 1 according to example embodiments.

[0039] Referring to FIG. 1, the substrate polishing apparatus 1 according to example embodiments may include a platen 10, a carrier head 20, a slurry supply member 30, and a pad conditioner 40.

[0040] The substrate polishing apparatus 1 performs a polishing process on a substrate. The substrate polishing apparatus 1 may perform the polishing process using a chemical mechanical method. The substrate on which the polishing process is performed may be a wafer or the like.

[0041] The platen 10 is provided to have a predetermined area. An upper surface of the platen 10 may have a flat plate shape. The upper surface of the platen 10 may have a larger area than that of the substrate, which may be a package substrate, and may be a polishing target. The platen 10 may have a circular flat plate structure. The platen 10 may be provided to be rotatable with respect to a central axis facing a vertical direction.

[0042] A polishing pad 11 is disposed above the upper surface of the platen 10. As an example, the polishing pad 11 may be attached to the upper surface of the platen 10. Accordingly, the polishing pad 11 may be rotated together with the platen 10. The polishing pad 11 may be provided as a flat plate with a predetermined thickness. The polishing pad 11 may contact the substrate to mechanically polish the substrate. A surface of the polishing pad 11 may be flannel, a paper material, a polyurethane material, a fiber material, or the like. Additionally, the surface of the polishing pad 11 may be in a state in which a diamond particle or the like is attached. A plurality of micropores may be formed at the polishing pad 11. The micropores formed at the polishing pad 11 may effectively accommodate slurry.

[0043] A platen driving member 16 may be connected to the platen 10. The platen driving member 16 provides power for the platen 10 to rotate. The platen driving member 16 may include a motor or the like.

[0044] The carrier head 20 may be a substrate head disposed adjacent to the platen 10. The carrier head 20 may be disposed above the platen 10. The carrier head 20 may hold the substrate, and may press the substrate against the platen 10 during the process. Additionally, the carrier head 20 may be rotated with reference to an axis facing a vertical direction. Accordingly, the carrier head 20 may be rotated while pressing the held substrate against the platen 10 while the polishing process is in progress. Additionally, the carrier

head 20 may move in a vertical direction or a direction crossing the vertical direction so that a position of the carrier head 20 is adjusted. Accordingly, the carrier head 20 may move to a position for unloading the substrate on which the polishing process has been performed, or may move to a position for loading the substrate on which the polishing process is to be performed.

[0045] A head driving member 26 may be connected to the carrier head 20. The head driving member 26 may provide power for rotating the carrier head 20 based on an axis facing a vertical direction. Additionally, the head driving member 26 may provide power for the carrier head 20 to move in a vertical direction or a direction crossing the vertical direction. For this purpose, the head driving member 26 may include a motor or the like.

[0046] The slurry supply member 30 may be disposed adjacent to the platen 10. The slurry supply member 30 supplies slurry in a direction of the platen 10. The slurry supply member 30 may supply the slurry to a surface of the polishing pad 11. The slurry may include a reactant, an abrasion particle, a chemical reaction catalyst, or the like.

[0047] A slurry storage member 36 may be connected to the slurry supply member 30. The slurry storage member 36 may store the slurry, and may supply the slurry to the slurry supply member 30.

[0048] A pad conditioner 40 may be disposed adjacent to the platen 10. The pad conditioner 40 may maintain a state of the polishing pad 11 so that the substrate is effectively polished. The pad conditioner 40 may be rotated with reference to an axis extending in a vertical direction. A rotation direction of the pad conditioner 40 may be the same as or different from a rotation direction of the platen 10. Additionally, the pad conditioner 40 may move in a vertical direction or a direction crossing the vertical direction so that a position of the pad conditioner 40 is adjusted. Accordingly, the pad conditioner 40 may move with respect to the platen 10 so that a sweep motion occurs with respect to the platen 10. A contact point between the pad conditioner 40 and the polishing pad 11 may be changed by the sweep motion.

[0049] The pad conditioner 40 may include a conditioning body 41 and a conditioning disk 42.

[0050] The conditioning body 41 has a predetermined volume. The conditioning body 41 may be rotated with reference to an axis facing a vertical direction. A rotation direction of the conditioning body 41 may be the same as or different from a rotation direction of the platen 10. Additionally, the conditioning body 41 may be provided to be movable in a vertical direction. Additionally, the conditioning body 41 may be provided to be movable in a direction crossing the vertical direction.

[0051] The conditioning disk 42 may be disposed on a bottom surface of the conditioning body 41. The conditioning disk 42 may be attached to or detached from the conditioning body 41.

[0052] A conditioner driving member 46 may be connected to the pad conditioner 40. The conditioner driving member 46 may provide power for rotating the pad conditioner 40 based on an axis extending in a vertical direction. Additionally, the conditioner driving member 46 may provide power for the pad conditioner 40 to move in a vertical direction or a direction crossing the vertical direction. To this end, the conditioner driving member 46 may include a motor or the like. FIG. 2 is a longitudinal cross-sectional view of the carrier head 20 of FIG. 1.

[0053] Referring to FIG. 2, the carrier head 20 may include a carrier body 21, a membrane 22, and a retainer ring 23.

[0054] The carrier body 21 has a predetermined volume. An outer circumference of the carrier body 21 may be a pillar shape or the like with a predetermined length in a vertical direction. The carrier body 21 may be rotated with reference to an axis extending in the vertical direction. Additionally, the carrier body 21 may be provided to be movable in the vertical direction. Additionally, the carrier body 21 may be provided to be movable in a direction crossing the vertical direction. A rotation direction of the carrier body 21 may be the same as or different from a rotation direction of the platen 10.

[0055] The membrane 22 may be disposed below a lower portion of the carrier body 21. An area of the membrane 22 along a direction crossing a vertical direction may correspond to the substrate. The membrane 22 may fix the substrate. As an example, the membrane 22 may form a negative pressure in at least some regions of a surface facing the substrate, so that the substrate is fixed through the negative pressure. In addition, the membrane 22 may be provided so that at least some regions of a surface facing the substrate expand. Thus, the membrane 22 may allow the substrate to be pressed toward the platen 10. The head driving member 26 may be connected to the membrane 22, so that the negative pressure is generated in the membrane 22 or the membrane 22 is expanded. To this end, the head driving member 26 may include a pump that discharges a gas from the membrane 22 or supplies a gas to the membrane 22.

[0056] FIG. 3 is an upper perspective view of the retainer ring 23 of FIG. 2, FIG. 4 is a lower perspective view of the retainer ring of FIG. 2, FIG. 5 is a partial plan view of the retainer ring 23 of FIG. 2, FIG. 6 is a longitudinal cross-sectional view of a lower outer side of the carrier head 20 passing through a region in which a weight adjusting groove 252 is disposed at the retainer ring 23, and FIG. 7 is a longitudinal cross-sectional view of a lower outer side of the carrier head 20 passing through a region in which a fastening groove 251a and 251b is disposed at the retainer ring 23.

[0057] Referring to FIGS. 3 to 7, the retainer ring 23 is disposed below the carrier head 20. The retainer ring 23 may be disposed at an outer region of a lower portion of the carrier head 20. The retainer ring 23 may be disposed at an outer region of a lower surface of the carrier body 21. A fastening portion 210 having a predetermined width in a radial direction may be disposed at an outer circumference of a lower portion of the carrier body 21. In addition, the retainer ring 23 may be disposed at the fastening portion 210. The retainer ring 23 may be disposed at an outer circumference of the membrane 22. The retainer ring 23 may be provided in a circular ring shape. According to example embodiments, the weight adjusting groove may have a ring shape. Accordingly, the retainer ring 23 may include an inner surface 230 facing an inner center C, an outer surface 240 disposed in an opposite direction to the inner surface 230 on a horizontal surface and facing an opposite side of the inner center C, and an upper surface 250 and a lower surface 260 disposed between the inner surface 230 and the outer surface 240. When the retainer ring 23 is coupled to the carrier head 20, the upper surface 250 faces the carrier head 20. Accordingly, the upper surface 250 of the retainer ring 23 may be a coupling surface or a surface facing the carrier head 20.

[0058] The inner surface 230 of the retainer ring 23 may be spaced apart from the membrane 22 by a predetermined distance. Accordingly, the membrane 22 may be prevented from being damaged by friction between the membrane 22 and the retainer ring 23. The inner surface 230 of the retainer ring 23 may have a structure in which a lower portion thereof protrudes toward the center C of the retainer ring 23 compared with an upper portion thereof.

[0059] The fastening groove 251a and 251b may be disposed at the upper surface 250 of the retainer ring 23. The fastening groove 251a and 251b may be recessed downward into the retainer ring from an upper surface of the retainer ring 23 to a predetermined depth. The fastening groove 251a and 251b may extend in a vertical direction, or may be inclined with respect to the vertical direction. A plurality of fastening grooves 251a and 251b may be configured to be spaced apart from each other along a circumferential direction. A distance between adjacent fastening grooves 251a and 251b may be the same or different.

[0060] Substrate carrier heads are provided in which a fastening groove recessed downward into the retainer ring by a predetermined depth is disposed at the upper surface of the retainer ring, and in which the carrier head further includes a fastening member that protrudes downward from the carrier body toward the upper surface of the retainer ring, the fastening member being configured to be inserted into the fastening groove.

[0061] The fastening groove may be one of a plurality of fastening grooves, wherein each of the fastening grooves is spaced apart from each other of the fastening grooves along the circumferential direction of the carrier head.

[0062] The fastening groove 251a and 251b may include a main fastening groove 251a and an auxiliary fastening groove 251b. At least one main fastening groove 251a may be provided. When viewed from above, the main fastening groove 251a may have a circular shape. At least one auxiliary fastening groove 251b may be provided. When viewed from above, the auxiliary fastening groove 251b may have a slit shape with a predetermined length. For example, when viewed from above, the auxiliary fastening groove 251b may have the slit shape with a predetermined length in a radial direction.

[0063] The retainer ring 23 may be coupled to the carrier body 21 through the fastening groove 251a and 251b. A fastening member 273 may protrude downward at a lower portion of the carrier body 21. The fastening member 273 may protrude to extend into the retainer ring 23. If the fastening member 273 is inserted into the fastening groove 251a and 251b, the retainer ring 23 may be coupled to the carrier body 21. As an example, the fastening member 273 may be a bolt, a pin, or the like inserted into the fastening portion 210 of the carrier body 21 to couple the retainer ring 23 to the carrier body 21. Additionally, the fastening member 273 may be a pin, a rib, or the like protruding downward from a lower surface of the fastening portion 210 to couple the retainer ring 23 to the carrier body 21.

[0064] The weight adjusting groove 252 may be disposed at the upper surface 250 of the retainer ring 23. The weight adjusting groove may be one of a plurality of weight adjusting grooves. A plurality of weight adjusting grooves 252 may be spaced apart from each other along a circumferential direction of the retainer ring. When viewed from above, an individual area of each weight adjusting groove 252 may be larger than an individual area of each fastening

groove 251a and 251b. The weight adjusting groove 252 may have a predetermined length along the circumferential direction. The weight adjusting groove 252 may have an arc shape. An end portion of the weight adjusting groove 252 along a length direction may be rounded. A length of each weight adjusting groove 252 along the circumferential direction may be greater than a length of each fastening groove 251a and 251b along the circumferential direction.

[0065] The weight adjusting groove 252 may have a predetermined width in a radial direction. The width of each weight adjusting groove 252 along the radial direction may be larger than a width of the main fastening groove 251a along a radial direction.

[0066] A weight adjusting groove 252 is provided to have a predetermined depth downward into the retainer ring, from the upper surface 250 of the retainer ring 23. A depth A1 of the weight adjusting groove 252 may be greater than 0 times and less than or equal to 0.8 times compared with a height A of the retainer ring 23.

[0067] An inner end of the weight adjusting groove 252 may be spaced apart a predetermined distance outward in a radial direction from an upper end of the inner surface of the retainer ring 23. An outer end of the weight adjusting groove 252 may be spaced apart a predetermined distance inward in the radial direction from an upper end of the outer surface of the retainer ring 23. At least one weight adjusting groove 252 may be disposed between adjacent fastening grooves 251a and 251b. A separation distance B between the weight adjusting groove 252 and the fastening groove 251a and 251b along the circumferential direction may be 2 mm or more. Accordingly, in a process of coupling the retainer ring 23 to the carrier body 21, the retainer ring 23 may be prevented from being damaged by the fastening member 273.

[0068] A volume of the weight adjusting groove 252 may be greater than 0 times and less than or equal to 1 time compared with a volume of the retainer ring 23. The volume of the weight adjusting groove 252 may be greater than 0 times and less than or equal to 0.5 times compared with a volume of the retainer ring 23 in a state in which the weight adjusting groove 252 is omitted. If a ratio of the volume of the weight adjusting groove 252 to the volume of the retainer ring 23 is greater than 1 time, there may be a problem in which rigidity of the retainer ring 23 is lowered and a possibility of damage in a use process increases.

[0069] A seal ring 271 and 272 may be disposed between the retainer ring 23 and the carrier body 21.

[0070] The seal ring 271 and 272 may include the inner seal ring 271 and the outer seal ring 272.

[0071] The inner seal ring 271 may be disposed at a region between the weight adjusting groove 252 and an upper end portion of the inner surface 230 of the retainer ring 23. The inner seal ring 271 may be disposed apart from the weight adjusting groove 252 by a predetermined distance inward in a radial direction. An inner seal ring insertion groove 253 may be disposed at the upper surface 250 of the retainer ring 23. By way of example, the seal ring may be disposed between the retainer ring and the carrier body.

[0072] The inner seal ring insertion groove 253 may be disposed in a ring shape on the upper surface 250 of the retainer ring 23. The inner seal ring insertion groove 253 may be disposed at the region between the weight adjusting groove 252 and the upper end portion of the inner surface 230 of the retainer ring 23. The inner seal ring insertion

groove 253 may be spaced apart from the weight adjusting groove 252 by a predetermined distance inward in a radial direction. At least a portion of the inner seal ring 271 may be configured to be inserted into the inner seal ring insertion groove 253.

[0073] An inner seal ring upper insertion groove 211 may be disposed at a lower surface of the carrier body 21. The inner seal ring upper insertion groove 211 may be disposed in a ring shape on the lower surface of the carrier body 21. The inner seal ring upper insertion groove 211 may be disposed at a region facing in a vertical direction the region between the weight adjusting groove 252 and the upper end portion of the inner surface 230 of the retainer ring 23. The inner seal ring upper insertion groove 211 may be disposed at a lower surface of the fastening portion 210. The inner seal ring upper insertion groove 211 may be disposed to face the inner seal ring insertion groove 253 in a vertical direction. At least a portion of the inner seal ring 271 may be configured to be inserted into the inner seal ring upper insertion groove 211.

[0074] The outer seal ring 272 may be disposed at a region between the weight adjusting groove 252 and an upper end portion of the outer surface 240 of the retainer ring 23. The outer seal ring 272 may be spaced apart from the weight adjusting groove 252 by a predetermined distance outward in a radial direction.

[0075] An outer seal ring insertion groove 254 may be disposed at the upper surface 250 of the retainer ring 23. The outer seal ring insertion groove 254 may be disposed in a ring shape on the upper surface 250 of the retainer ring 23. The outer seal ring insertion groove 254 may be disposed at the region between the weight adjusting groove 252 and the upper end portion of the outer surface 240 of the retainer ring 23. The outer seal ring insertion groove 254 may be spaced apart from the weight adjusting groove 252 by a predetermined distance outward in a radial direction. At least a portion of the outer seal ring 272 may be configured to be inserted into the outer seal ring insertion groove 254.

[0076] An outer seal ring upper insertion groove 212 may be disposed at a lower surface of the carrier body 21. The outer seal ring upper insertion groove 212 may be disposed in a ring shape on the lower surface of the carrier body 21. The outer seal ring upper insertion groove 212 may be disposed at a region facing in a vertical direction the region between the weight adjusting groove 252 and the upper end portion of the outer surface 240 of the retainer ring 23. The outer seal ring upper insertion groove 212 may be disposed at a lower surface of the fastening portion 210. The outer seal ring upper insertion groove 212 may be disposed to face the outer seal ring insertion groove 254 in a vertical direction. At least a portion of the outer seal ring 272 may be configured to be inserted into the outer seal ring upper insertion groove 212.

[0077] A supply path 261 may be disposed at the lower surface 260 of the retainer ring 23. The supply path 261 may be recessed upward at the lower surface 260 of the retainer ring 23. An inner end of the supply path 261 is connected to a space disposed inside the retainer ring 23, and an outer end of the supply path 261 is connected to a space disposed outside the retainer ring 23. Accordingly, the space disposed inside the retainer ring 23 and the space disposed outside the retainer ring 23 are connected to each other by the supply path 261. A plurality of supply paths 261 may be configured to be spaced apart from each other along a circumferential

direction. A length direction of the supply path 261 may be provided toward a radial direction, or may be provided inclined to the radial direction. Accordingly, the slurry may flow between the space disposed inside the retainer ring 23 and the space disposed outside the retainer ring 23 through the supply path 261.

[0078] The outer surface 240 of the retainer ring 23 may have a structure in which a lower portion thereof is recessed toward the center C of the retainer ring 23 compared with an upper portion thereof. Accordingly, a step difference surface 241 extending from the outside toward the inside in a radial direction may be disposed at the outer surface 240 of the retainer ring 23.

[0079] A height A3 from an upper end of the retainer ring 23 to an outer end of the step difference surface 241 along the radial direction may be 0.5 times or more and 0.8 times or less compared with the height A of the retainer ring 23. Accordingly, a height A4 from a lower end of the retainer ring 23 to the outer end of the step difference surface 241 along the radial direction may be 0.2 to 0.5 times the height A of the retainer ring 23. The depth A1 of the weight adjusting groove 252 may be less than or equal to the height A3 from the upper end of the retainer ring 23 to the outer end of the step difference surface 241 along the radial direction.

[0080] The retainer ring 23 may include an upper ring 24 and a lower ring 25.

[0081] The upper ring 24 is disposed at an upper region of the retainer ring 23. The upper ring 24 may be coupled to the carrier body 21. The upper ring 24 may be made of a metal material. The fastening groove 251a and 251b, the weight adjusting groove 252, the inner seal ring insertion groove 253, and the outer seal ring insertion groove 254 may be disposed at the upper ring 24.

[0082] An insertion portion 270 recessed from a lower side toward an upper side may be disposed at a lower portion of the upper ring 24. The insertion portion 270 may be provided in a ring shape along a circumferential direction. For example, a lower surface of the upper ring 24 may have a structure in which an inner region of the insertion portion 270 along a radial direction protrudes downward compared with an outer region of the insertion portion 270 along the radial direction. The lower surface of the upper ring 24 may have a structure in which a step difference is formed so that the inner region of the insertion portion 270 along the radial direction protrudes downward.

[0083] Additionally, the insertion portion 270 may have a groove structure recessed from a lower side toward an upper side compared with a region adjacent to the inner region of the insertion portion 270 along the radial direction and a region adjacent to the outer region of the insertion portion 270 along the radial direction.

[0084] Additionally, the lower surface of the upper ring 24 may have a structure in which an outer region of the insertion portion 270 along a radial direction protrudes downward compared with an inner region of the insertion portion 270 along the radial direction. The lower surface of the upper ring 24 may have a structure in which a step difference is formed so that the outer region of the insertion portion 270 along the radial direction protrudes downward.

[0085] In FIGS. 2, 6, and 7, a case in which the lower surface of the upper ring 24 has a structure in which the step difference is formed so that the inner region of the insertion portion 270 along the radial direction protrudes downward is illustrated.

[0086] The lower ring 25 is disposed at a lower region of the retainer ring 23. The lower ring 25 may be coupled to the upper ring 24. As an example, the lower ring 25 may be coupled to the upper ring 24 in a method in which it is bonded through an adhesive or the like. A material of the lower ring 25 may be different from a material of the upper ring 24. The material of the lower ring 25 may have a lower hardness than that of the material of the upper ring 24. The lower ring 25 may be made of a resin material. The supply path 261 may be disposed at the lower ring 25. An assembly protrusion (or a protruding portion for assembly) 276 protruding from a lower side toward an upper side may be disposed at an upper portion of the lower ring 25. The assembly protrusion 276 may be provided in a ring shape along a circumferential direction. As an example, the assembly protrusion 276 may be disposed at an outer region of the lower ring 25 along a radial direction. Accordingly, an upper surface of the lower ring 25 may have a structure in which an inner region of the lower ring 25 along the radial direction protrudes upward compared with the outer region of the lower ring 25 along the radial direction. The upper surface of the lower ring 25 may have a structure in which a step difference is formed so that the outer region of the lower ring 25 along the radial direction protrudes upward.

[0087] Additionally, the assembly protrusion 276 may have a protrusion structure protruding upward compared with a region adjacent to the inner region of the lower ring 25 along the radial direction and a region adjacent to the outer region of the lower ring 25 along the radial direction.

[0088] Additionally, the assembly protrusion 276 may be disposed at the inner region of the lower ring 25 along the radial direction. Accordingly, the upper surface of the lower ring 25 may have a structure in which the outer region of the lower ring 25 along the radial direction protrudes upward compared with the inner region of the lower ring 25 along the radial direction. The upper surface of the lower ring 25 may have a structure in which a step difference is formed so that the inner region of the lower ring 25 along the radial direction protrudes upward.

[0089] In FIGS. 2, 6, and 7, a case in which the assembly protrusion 276 is disposed at the outer region of the lower ring 25 along the radial direction is illustrated.

[0090] The lower ring 25 may have a structure in which the assembly protrusion 276 is inserted into the insertion portion 270, and may be aligned below the upper ring 24. An outer end of a lower surface of the upper ring 24 along a radial direction may be the step difference surface 241. In addition, the weight adjusting groove 252 may be disposed only at the upper ring 24.

[0091] A weight of the retainer ring 23 affects a polishing state of an edge region of the substrate. Specifically, a phenomenon in which the polishing pad 11 is pressed by the retainer ring 23 may occur. Accordingly, a portion of the polishing pad 11 that is pressed by the retainer ring 23 may be concave downward, and a region adjacent to the concave portion may be convex upward. Accordingly, the polishing pad 11 is curved at a region adjacent to a portion in contact with the retainer ring 23 and a portion in contact with the retainer ring 23, and a curvature of the polishing pad 11 affects the polishing state of the edge region of the substrate.

[0092] Accordingly, it is necessary to adjust a weight of the retainer ring 23 in order to adjust the polishing state of the edge region of the substrate. However, if the weight of the retainer ring 23 is changed, a shape of the retainer ring

23 changes as well. If the shape of the retainer ring 23 changes, a flow form of slurry flowing between a space disposed inside the retainer ring 23 and a space disposed outside the retainer ring 23 and a shape of a curvature occurring at the polishing pad 11 also change. Accordingly, changing of the retainer ring 23 requires complex adjustment of various design elements, so that there is a problem in which it is difficult to change the weight of the retainer ring 23.

[0093] In the substrate polishing apparatus 1 according to the embodiment, the weight of the retainer ring 23 may be adjusted by the weight adjusting groove 252. Additionally, the weight adjusting groove 252 is disposed at an upper surface 250 of the retainer ring 23. Thus, if the retainer ring 23 is coupled to the carrier body 21, the weight adjusting groove 252 is not exposed to the outside. In other words, a shape of the retainer ring 23 exposed to the outside does not change depending on presence or absence of the weight adjusting groove 252. Accordingly, in the substrate polishing apparatus 1 according to the embodiment, the weight of the retainer ring 23 may be adjusted while a change in another design element is minimized.

[0094] Additionally, in the substrate polishing apparatus 1 according to the embodiment, the seal ring 271 and 272 may be disposed between the retainer ring 23 and the carrier body 21. Accordingly, slurry or the like may be prevented from being introduced between the retainer ring 23 and the carrier body 21.

[0095] FIG. 8 is a cross-sectional view of a retainer ring 23a according to another embodiment.

[0096] Referring to FIG. 8, a supplementary member 280 may be disposed inside a weight adjusting groove 252a. The supplementary member 280 may fill an entire weight adjusting groove 252a, or may fill a portion of the weight adjusting groove 252a. A density of the supplementary member 280 may be greater than a density of the retainer ring 23a, may be less than the density of the retainer ring 23a, or may be the same as the density of the retainer ring 23a. If the retainer ring 23a is made of two or more different materials, the density of the retainer ring 23a may be an average density.

[0097] According to non-limiting examples, a supplementary member may be disposed inside a weight adjusting groove. For example, the retainer ring may include an upper ring disposed at an upper region of the retainer ring and a lower ring disposed at a lower region of the retainer ring, wherein the weight adjusting groove is disposed only at the upper ring.

[0098] According to further embodiments, the retainer ring includes a plurality of weight adjusting grooves, wherein the plurality of weight adjusting grooves are configured to be spaced apart from each other along a circumferential direction of the retainer ring.

[0099] The weight adjusting groove has a ring shape.

[0100] Because a structure of the retainer ring 23a according to the embodiment of FIG. 8 may be the same as or similar to that of the retainer ring 23 described above in FIGS. 2 to 7 except that the supplementary member 280 is disposed in the weight adjusting groove 252a, a repeated description of corresponding elements thereof is omitted.

[0101] In the retainer ring 23a according to the embodiment of FIG. 8 may be, the supplementary member 280 may be additionally filled in the weight adjusting groove 252a. Accordingly, after the weight adjusting groove 252a is

formed in the retainer ring 23a, rigidity of the retainer ring 23a may be supplemented according to the supplementary member 280 filled in the weight adjusting groove 252a, a total weight of the retainer ring 23a may be adjusted according to the supplementary member 280 filled in the weight adjusting groove 252a, or a mass distribution along a circumferential direction of the retainer ring 23a may be additionally adjusted according to the supplementary member 280 filled in the weight adjusting groove 252a.

[0102] FIG. 9 is a cross-sectional view of a retainer ring 23b according to another embodiment.

[0103] Referring to FIG. 9, upper and lower portions of the retainer ring 23b may be made of the same material. As an example, an entire region of the retainer ring 23b exposed to the outside may be provided as a single material.

[0104] In addition, because a structure of the retainer ring 23b is the same as or similar to that of the retainer ring 23 described above in FIGS. 2 to 7 or that of the retainer ring 23a described above in FIG. 8, a repeated description of corresponding elements thereof is omitted.

[0105] FIG. 10 is a view showing a disposition structure of a seal ring 271c and 272c according to another embodiment.

[0106] Referring to FIG. 10, the seal ring 271c and 272c may be disposed between a retainer ring 23c and a carrier body 21c.

[0107] The seal ring 271c and 272c may include an inner seal ring 271c and an outer seal ring 272c.

[0108] The inner seal ring 271c may be disposed at a region between a weight adjusting groove 252c and an upper end portion of an inner surface of the retainer ring 23c. The inner seal ring 271c may be configured to be spaced apart from the weight adjusting groove 252c by a predetermined distance inward in a radial direction.

[0109] An inner seal ring insertion groove 253c may be disposed at an upper surface of the retainer ring 23c. Because the inner seal ring insertion groove 253c is the same as or similar to the inner seal ring insertion groove 253 of the retainer ring 23 described above in FIGS. 2 to 7, a repeated description of corresponding elements thereof is omitted.

[0110] At least a portion of the inner seal ring 271c may be configured to be inserted into the inner seal ring insertion groove 253c.

[0111] In addition, the inner seal ring upper insertion groove 211 described above in FIGS. 2 to 7 may be omitted from the carrier body 21c.

[0112] The outer seal ring 272c may be disposed at a region between the weight adjusting groove 252c and an upper end portion of an outer surface of the retainer ring 23c. The outer seal ring 272c may be spaced apart from the weight adjusting groove 252c by a predetermined distance outward in a radial direction.

[0113] An outer seal ring insertion groove 254c may be disposed at an upper surface of the retainer ring 23c. Because the outer seal ring insertion groove 254c is the same as or similar to the outer seal ring insertion groove 254 of the retainer ring 23 described above in FIGS. 2 to 6, a repeated description of corresponding elements is omitted.

[0114] At least a portion of the outer seal ring 272c may be configured to be inserted into the outer seal ring insertion groove 254c.

[0115] In addition, the outer seal ring upper insertion groove 212 may be as described above in FIGS. 2 to 7 may be omitted from the carrier body 21c.

[0116] In addition, because a structure of the retainer ring 23c is the same as or similar to that of the retainer ring 23 described above in FIGS. 2 to 7, that of the retainer ring 23a described above in FIG. 8, or that of the retainer ring 23b described above in FIG. 9, a repeated description of corresponding elements thereof is omitted.

[0117] Additionally, one of the inner seal ring 271c and the outer seal ring 272c may be disposed as a disposition structure of the seal ring 271 and 272 described above in FIGS. 2 to 7.

[0118] FIG. 11 is a view showing a disposition structure of a seal ring 271d and 272d according to another embodiment.

[0119] Referring to FIG. 11, the seal ring 271d and 272d may be disposed between a retainer ring 23d and a carrier body 21d.

[0120] The seal ring 271d and 272d may include the inner seal ring 271d and the outer seal ring 272d.

[0121] The inner seal ring 271d may be disposed at a region between a weight adjusting groove 252d and an upper end portion of an inner surface of the retainer ring 23d. The inner seal ring 271d may be spaced apart from the weight adjusting groove 252d by a predetermined distance inward in a radial direction.

[0122] An inner seal ring upper insertion groove 211d may be disposed at a lower surface of the carrier body 21d. Because the inner seal ring upper insertion groove 211d is the same as or similar to the inner seal ring upper insertion groove 211 of the carrier body 21 described above in FIGS. 2 to 7, a repeated description of corresponding elements thereof is omitted.

[0123] At least a portion of the inner seal ring 271d may be configured to be inserted into the inner seal ring upper insertion groove 211d.

[0124] Additionally, the inner seal ring insertion groove 253 described above in FIGS. 2 to 7 may be omitted from the retainer ring 23d.

[0125] The outer seal ring 272d may be disposed at a region between the weight adjusting groove 252d and an upper end portion of an outer surface of the retainer ring 23d. The outer seal ring 272d may be configured to be spaced apart from the weight adjusting groove 252d by a predetermined distance outward in a radial direction.

[0126] An outer seal ring upper insertion groove 212d may be disposed at a lower surface of the carrier body 21d. Because the outer seal ring upper insertion groove 212d is the same as or similar to the outer seal ring upper insertion groove 212 of the carrier body 21 described above in FIGS. 2 to 7, a repeated description of corresponding elements thereof is omitted.

[0127] At least a portion of the outer seal ring 272d may be configured to be inserted into the outer seal ring upper insertion groove 212d.

[0128] Additionally, the outer seal ring insertion groove 254 described above in FIGS. 2 to 7 may be omitted from the retainer ring 23d.

[0129] In addition, because a structure of the retainer ring 23d is the same as or similar to that of the retainer ring 23 described above in FIGS. 2 to 7, that of the retainer ring 23a described above in FIG. 8, or that of the retainer ring 23b described above in FIG. 9, a repeated description of corresponding elements thereof is omitted.

[0130] Additionally, one of the inner seal ring 271d and the outer seal ring 272d may be disposed as a disposition structure of the seal ring 271 and 272 described above in

FIGS. 2 to 7 or a disposition structure of the seal ring 271c and 272c described above in FIG. 10.

[0131] FIG. 12 is a view showing a disposition structure of a seal ring 271e and 272e according to another embodiment.

[0132] Referring to FIG. 12, the seal ring 271e and 272e may be disposed between a retainer ring 23e and a carrier body 21e.

[0133] The seal ring 271e and 272e may include the inner seal ring 271e and the outer seal ring 272e.

[0134] The inner seal ring 271e may be disposed at a region between a weight adjusting groove 252e and an upper end portion of an inner surface of the retainer ring 23e. The inner seal ring 271e may be configured to be spaced apart from the weight adjusting groove 252e by a predetermined distance inward in a radial direction. The inner seal ring 271e is disposed so that some regions thereof are exposed to a space disposed inside the retainer ring 23e. The inner seal ring 271e may be inserted outward in a radial direction in the space disposed inside the retainer ring 23e, and may be disposed between the retainer ring 23e and the carrier body 21e.

[0135] An inner seal ring insertion groove 253e may be disposed at an upper portion of the retainer ring 23e. The inner seal ring insertion groove 253e is provided as a groove structure recessed into the retainer ring from the upper end portion of the inner surface of the retainer ring 23e toward an outer side of the retainer ring 23e along a radial direction. The inner seal ring insertion groove 253e may have a different depth recessed toward the outer side along the radial direction for each region along a vertical direction. A depth recessed toward the outer side along the radial direction of a lower region of the inner seal ring insertion groove 253e may be greater than that of an upper region of the inner seal ring insertion groove 253e. The inner seal ring insertion groove 253e may be disposed in a ring shape at the upper end portion of the inner surface of the retainer ring 23e. The inner seal ring insertion groove 253e may be disposed at a region between the weight adjusting groove 252e and the upper end portion of the inner surface of the retainer ring 23e. The inner seal ring insertion groove 253e may be spaced apart from the weight adjusting groove 252e by a predetermined distance inward in a radial direction.

[0136] A lower portion of the inner seal ring 271e has a shape corresponding to the inner seal ring insertion groove 253e, and is disposed in the inner seal ring insertion groove 253e.

[0137] An inner seal ring upper insertion groove 211e may be disposed at a lower surface of the carrier body 21e. The inner seal ring upper insertion groove 211e may be provided as a groove structure recessed from a lower end portion of an inner surface of a fastening portion 210e toward an outer side of the fastening portion 210e along a radial direction. The inner seal ring insertion groove 253e may have a different depth recessed into the retainer ring toward the outer side along the radial direction for each region along a vertical direction. A depth recessed into the retainer ring toward the outer side along the radial direction of an upper region of the inner seal ring upper insertion groove 211e may be greater than that of a lower region of the inner seal ring upper insertion groove 211e.

[0138] The inner seal ring upper insertion groove 211e may be disposed in a ring shape at a lower end portion of the fastening portion 210e.

[0139] The inner seal ring upper insertion groove 211e may be disposed at a region facing in a vertical direction the region between the weight adjusting groove 252e and the upper end portion of the inner surface of the retainer ring 23e. The inner seal ring upper insertion groove 211e may be disposed to face the inner seal ring insertion groove 253e in a vertical direction.

[0140] An upper portion of the inner seal ring 271e has a shape corresponding to the inner seal ring upper insertion groove 211e, and is disposed in the inner seal ring upper insertion groove 211e.

[0141] The outer seal ring 272e may be disposed at a region between the weight adjusting groove 252e and an upper end portion of an outer surface of the retainer ring 23e. The outer seal ring 272e may be spaced apart from the weight adjusting groove 252e by a predetermined distance outward in a radial direction. The outer seal ring 272e is disposed so that some regions thereof are exposed to a space disposed outside the retainer ring 23e. The outer seal ring 272e may be inserted inward in a radial direction in the space disposed outside the retainer ring 23e, and may be disposed between the retainer ring 23e and the carrier body 21e.

[0142] An outer seal ring insertion groove 254e may be disposed at an upper portion of the retainer ring 23e. The outer seal ring insertion groove 254e is provided as a groove structure recessed into the retainer ring from the upper end portion of the outer surface of the retainer ring 23e toward an inner side of the retainer ring 23e along a radial direction. The outer seal ring insertion groove 254e may have a different depth recessed into the retainer ring toward the inner side along the radial direction for each region along a vertical direction. A depth recessed into the retainer ring toward the inner side along the radial direction of a lower region of the outer seal ring insertion groove 254e may be greater than that of an upper region of the outer seal ring insertion groove 254e. The outer seal ring insertion groove 254e may be disposed in a ring shape at the upper end portion of the outer surface of the retainer ring 23e. The outer seal ring insertion groove 254e may be disposed at a region between the weight adjusting groove 252e and the upper end portion of the outer surface of the retainer ring 23e. The outer seal ring insertion groove 254e may be spaced apart from the weight adjusting groove 252e by a predetermined distance outward in a radial direction.

[0143] A lower portion of the outer seal ring 272e has a shape corresponding to the outer seal ring insertion groove 254e, and is disposed in the outer seal ring insertion groove 254e.

[0144] An outer seal ring upper insertion groove 212e may be disposed at a lower surface of the carrier body 21e. The outer seal ring upper insertion groove 212e may be provided as a groove structure recessed into the retainer ring from a lower end portion of an outer surface of the fastening portion 210e toward an inner side of the fastening portion 210e along a radial direction. The outer seal ring upper insertion groove 212e may have a different depth recessed into the retainer ring toward the inner side along the radial direction for each region along a vertical direction. A depth recessed into the retainer ring toward the inner side along the radial direction of an upper region of the outer seal ring upper insertion groove 212e may be greater than that of a lower region of the outer seal ring upper insertion groove 212e.

The outer seal ring upper insertion groove **212e** may be disposed in a ring shape at a lower end portion of the fastening portion **210e**.

[0145] The outer seal ring upper insertion groove **212e** may be disposed at a region facing in a vertical direction the region between the weight adjusting groove **252e** and the upper end portion of the outer surface of the retainer ring **23e**. The outer seal ring upper insertion groove **212e** may be disposed to face the outer seal ring insertion groove **254e** in a vertical direction.

[0146] An upper portion of the outer seal ring **272e** has a shape corresponding to the outer seal ring upper insertion groove **212e**, and is disposed in the outer seal ring upper insertion groove **212e**.

[0147] In addition, because a structure of the retainer ring **23e** is the same as or similar to that of the retainer ring **23** described above in FIGS. 2 to 7, that of the retainer ring **23a** described above in FIG. 8, or that of the retainer ring **23b** described above in FIG. 9, a repeated description of corresponding elements thereof is omitted.

[0148] Additionally, one of the inner seal ring **271e** and the outer seal ring **272e** may be disposed as a disposition structure of the seal ring **271** and **272** described above in FIGS. 2 to 7, a disposition structure of the seal ring **271c** and **272c** described above in FIG. 10, or a disposition structure of the seal ring **271d** and **272d** described above in FIG. 11.

[0149] FIG. 13 is a view showing a disposition structure of a seal ring **271f** and **272f** according to another embodiment.

[0150] Referring to FIG. 13, the seal ring **271f** and **272f** may be disposed between a retainer ring **23f** and a carrier body **21f**.

[0151] The seal ring **271f** and **272f** may include the inner seal ring **271f** and the outer seal ring **272f**.

[0152] The inner seal ring **271f** may be disposed at a region between a weight adjusting groove **252f** and an upper end portion of an inner surface of the retainer ring **23f**. The inner seal ring **271f** may be spaced apart from the weight adjusting groove **252f** by a predetermined distance inward in a radial direction. The inner seal ring **271f** is disposed so that some regions thereof are exposed to a space disposed inside the retainer ring **23f**. The inner seal ring **271f** may be inserted outward in a radial direction in the space disposed inside the retainer ring **23f**, and may be disposed between the retainer ring **23f** and the carrier body **21f**.

[0153] An inner seal ring insertion groove **253f** may be disposed at an upper portion of the retainer ring **23f**. The inner seal ring insertion groove **253f** is provided as a groove structure recessed into the retainer ring from the upper end portion of the inner surface of the retainer ring **23f** toward an outer side of the retainer ring **23f** along a radial direction. The inner seal ring insertion groove **253f** may be disposed in a ring shape at the upper end portion of the inner surface of the retainer ring **23f**. The inner seal ring insertion groove **253f** may be disposed at a region between the weight adjusting groove **252f** and the upper end portion of the inner surface of the retainer ring **23f**. The inner seal ring insertion groove **253f** may be spaced apart from the weight adjusting groove **252f** by a predetermined distance inward in a radial direction.

[0154] A lower portion of the inner seal ring **271f** has a shape corresponding to the inner seal ring insertion groove **253f**, and is disposed in the inner seal ring insertion groove **253f**.

[0155] An inner seal ring upper insertion groove **211f** may be disposed at a lower surface of the carrier body **21f**. Because the inner seal ring upper insertion groove **211f** is the same as or similar to the inner seal ring upper insertion groove **211** of the carrier body **21** described above in FIGS. 2 to 7, a repeated description of corresponding elements thereof is omitted.

[0156] An upper portion of the inner seal ring **271f** has a shape corresponding to the inner seal ring upper insertion groove **211f**, and is disposed in the inner seal ring upper insertion groove **211f**.

[0157] The outer seal ring **272f** may be disposed at a region between the weight adjusting groove **252f** and an upper end portion of an outer surface of the retainer ring **23f**. The outer seal ring **272f** may be spaced apart from the weight adjusting groove **252f** by a predetermined distance outward in a radial direction. The outer seal ring **272f** is disposed so that some regions thereof are exposed to a space disposed outside the retainer ring **23f**. The outer seal ring **272f** may be inserted inward in a radial direction in the space disposed outside the retainer ring **23f**, and may be disposed between the retainer ring **23f** and the carrier body **21f**.

[0158] An outer seal ring insertion groove **254f** may be disposed at an upper portion of the retainer ring **23f**. The outer seal ring insertion groove **254f** is provided as a groove structure recessed into the retainer ring from the upper end portion of the outer surface of the retainer ring **23f** toward an inner side of the retainer ring **23f** along a radial direction.

[0159] The outer seal ring insertion groove **254f** may be disposed in a ring shape at the upper end portion of the outer surface of the retainer ring **23f**. The outer seal ring insertion groove **254f** may be disposed at a region between the weight adjusting groove **252f** and the upper end portion of the outer surface of the retainer ring **23f**. The outer seal ring insertion groove **254f** may be spaced apart from the weight adjusting groove **252f** by a predetermined distance outward in a radial direction.

[0160] A lower portion of the outer seal ring **272f** has a shape corresponding to the outer seal ring insertion groove **254f**, and is disposed in the outer seal ring insertion groove **254f**.

[0161] An outer seal ring upper insertion groove **212f** may be disposed at a lower surface of the carrier body **21f**. Because the outer seal ring upper insertion groove **212f** is the same as or similar to the outer seal ring upper insertion groove **212** of the carrier body **21** described above in FIGS. 2 to 7, a repeated description of corresponding elements thereof is omitted.

[0162] An upper portion of the outer seal ring **272f** has a shape corresponding to the outer seal ring upper insertion groove **212f**, and is disposed in the outer seal ring upper insertion groove **212f**.

[0163] In addition, because a structure of the retainer ring **23f** is the same as or similar to that of the retainer ring **23** described above in FIGS. 2 to 7, that of the retainer ring **23a** described above in FIG. 8, or that of the retainer ring **23b** described above in FIG. 9, a repeated description of corresponding elements thereof is omitted.

[0164] Additionally, one of the inner seal ring **271f** and the outer seal ring **272f** may be disposed as a disposition structure of the seal ring **271** and **272** described above in FIGS. 2 to 7, a disposition structure of the seal ring **271c** and **272c** described above in FIG. 10, a disposition structure of

the seal ring 271*d* and 272*d* described above in FIG. 11, or a disposition structure of the seal ring 271*e* and 272*e* described above in FIG. 12.

[0165] FIG. 14 is a view showing a disposition structure of a seal ring 271*g* and 272*g* according to another embodiment.

[0166] Referring to FIG. 14, the seal ring 271*g* and 272*g* may be disposed between a retainer ring 23*g* and a carrier body 21*g*.

[0167] The seal ring 271*g* and 272*g* may include the inner seal ring 271*g* and the outer seal ring 272*g*.

[0168] The inner seal ring 271*g* may be disposed at a region between a weight adjusting groove 252*g* and an upper end portion of an inner surface of the retainer ring 23*g*. The inner seal ring 271*g* may be spaced apart from the weight adjusting groove 252*g* by a predetermined distance inward in a radial direction. The inner seal ring 271*g* is disposed so that some regions thereof are exposed to a space disposed inside the retainer ring 23*g*. The inner seal ring 271*g* may be inserted outward in a radial direction in the space disposed inside the retainer ring 23*g*, and may be disposed between the retainer ring 23*g* and the carrier body 21*g*.

[0169] An inner seal ring insertion groove 253*g* may be disposed at an upper portion of the retainer ring 23*g*. The inner seal ring insertion groove 253*g* is provided as a groove structure recessed into the retainer ring from the upper end portion of the inner surface of the retainer ring 23*g* toward an outer side of the retainer ring 23*g* along a radial direction. The inner seal ring insertion groove 253*g* may have a different depth recessed into the retainer ring toward the outer side along the radial direction for each region along a vertical direction. A depth recessed into the retainer ring toward the outer side along the radial direction of a lower region of the inner seal ring insertion groove 253*g* may be greater than that of an upper region of the inner seal ring insertion groove 253*g*. The inner seal ring insertion groove 253*g* may be disposed in a ring shape at the upper end portion of the inner surface of the retainer ring 23*g*. The inner seal ring insertion groove 253*g* may be disposed at the region between the weight adjusting groove 252*g* and the upper end portion of the inner surface of the retainer ring 23*g*. The inner seal ring insertion groove 253*g* may be spaced apart from the weight adjusting groove 252*g* by a predetermined distance inward in a radial direction.

[0170] A lower portion of the inner seal ring 271*g* has a shape corresponding to the inner seal ring insertion groove 253*g*, and is disposed in the inner seal ring insertion groove 253*g*.

[0171] An inner seal ring upper insertion groove 211*g* may be disposed at a lower surface of the carrier body 21*g*. Because the inner seal ring upper insertion groove 211*g* is the same as or similar to the inner seal ring upper insertion groove 211 of the carrier body 21 described above in FIGS. 2 to 7, a repeated description of corresponding elements thereof is omitted.

[0172] An upper portion of the inner seal ring 271*g* has a shape corresponding to the inner seal ring upper insertion groove 211*g*, and is disposed in the inner seal ring upper insertion groove 211*g*.

[0173] The outer seal ring 272*g* may be disposed at a region between the weight adjusting groove 252*g* and an upper end portion of an outer surface of the retainer ring 23*g*. The outer seal ring 272*g* may be spaced apart from the weight adjusting groove 252*g* by a predetermined distance outward in a radial direction. The outer seal ring 272*g* is

disposed so that some regions thereof are exposed to a space disposed outside the retainer ring 23*g*. The outer seal ring 272*g* may be inserted inward in a radial direction in the space disposed outside the retainer ring 23*g*, and may be disposed between the retainer ring 23*g* and the carrier body 21*g*.

[0174] An outer seal ring insertion groove 254*g* may be disposed at an upper portion of the retainer ring 23*g*. The outer seal ring insertion groove 254*g* is provided as a groove structure recessed into the retainer ring from the upper end portion of the outer surface of the retainer ring 23*g* toward an inner side of the retainer ring 23*g* along a radial direction. The outer seal ring insertion groove 254*g* may have a different depth recessed into the retainer ring toward the inner side along the radial direction for each region along a vertical direction. A depth recessed into the retainer ring toward the inner side along the radial direction of a lower region of the outer seal ring insertion groove 254*g* may be greater than that of an upper region of the outer seal ring insertion groove 254*g*. The outer seal ring insertion groove 254*g* may be disposed in a ring shape at the upper end portion of the outer surface of the retainer ring 23*g*. The outer seal ring insertion groove 254*g* may be disposed at the region between the weight adjusting groove 252*g* and the upper end portion of the outer surface of the retainer ring 23*g*. The outer seal ring insertion groove 254*g* may be configured to be spaced apart from the weight adjusting groove 252*g* by a predetermined distance outward in a radial direction.

[0175] A lower portion of the outer seal ring 272*g* has a shape corresponding to the outer seal ring insertion groove 254*g*, and is disposed in the outer seal ring insertion groove 254*g*.

[0176] An outer seal ring upper insertion groove 212*g* may be disposed at a lower surface of the carrier body 21*g*. Because the outer seal ring upper insertion groove 212*g* is the same as or similar to the outer seal ring upper insertion groove 212 of the carrier body 21 described above in FIGS. 2 to 7, a repeated description of corresponding elements thereof is omitted.

[0177] An upper portion of the outer seal ring 272*g* has a shape corresponding to the outer seal ring upper insertion groove 212*g*, and is disposed in the outer seal ring upper insertion groove 212*g*.

[0178] In addition, because a structure of the retainer ring 23*g* is the same as or similar to that of the retainer ring 23 described above in FIGS. 2 to 7, that of the retainer ring 23*a* described above in FIG. 8, or that of the retainer ring 23*b* described above in FIG. 9, a repeated description of corresponding elements thereof is omitted.

[0179] Additionally, one of the inner seal ring 271*g* and the outer seal ring 272*g* may be disposed as a disposition structure of the seal ring 271 and 272 described above in FIGS. 2 to 7, a disposition structure of the seal ring 271*c* and 272*c* described above in FIG. 10, a disposition structure of the seal ring 271*d* and 272*d* described above in FIG. 11, a disposition structure of the seal ring 271*e* and 272*e* described above in FIG. 12, or a disposition structure of the seal ring 271*f* and 272*f* described above in FIG. 13.

[0180] FIG. 15 is an upper perspective view of a retainer ring 23*h* according to another embodiment.

[0181] Referring to FIG. 15, a weight adjusting groove 252*h* may be disposed at an upper surface of the retainer ring 23*h*. The weight adjusting groove 252*h* may have a ring

shape. The weight adjusting groove **252h** may include an inner weight adjusting groove **291h** and an outer weight adjusting groove **292h**.

[0182] The inner weight adjusting groove **291h** may be disposed at an inner region of the upper surface of the retainer ring **23h** along a radial direction. An inner end of the inner weight adjusting groove **291h** may be spaced apart from an upper end of an inner surface of the retainer ring **23h** by a predetermined distance outward in a radial direction. The inner weight adjusting groove **291h** may be disposed inside the radial direction with respect to a main fastening groove **2511h**. An outer end of the inner weight adjusting groove **291h** may be spaced apart from the main fastening groove **2511h**. The inner weight adjusting groove **291h** may be connected to one end of an auxiliary fastening groove **2512h** along a length direction. The inner end of the inner weight adjusting groove **291h** may be spaced apart from an inner seal ring insertion groove **253h**. Accordingly, the inner seal ring insertion groove **253h** may be disposed at a region between the inner weight adjusting groove **291h** and an upper end portion of an inner surface of the retainer ring **23h**.

[0183] According to example embodiments, the retainer ring includes a plurality of weight adjusting grooves including an inner weight adjusting groove disposed at an inner region of the upper surface of the retainer ring along a radial direction; and an outer weight adjusting groove disposed at an outer region of the upper surface of the retainer ring along the radial direction. According to non-limiting examples of these and other embodiments, the weight adjusting groove may have a ring shape.

[0184] The outer weight adjusting groove **292h** may be disposed at an outer region of the upper surface of the retainer ring **23h** along a radial direction. An outer end of the outer weight adjusting groove **292h** may be spaced apart from an upper end of an outer surface of the retainer ring **23h** by a predetermined distance inward in a radial direction. The outer weight adjusting groove **292h** may be disposed outside the radial direction with respect to the main fastening groove **2511h**. An inner end of the outer weight adjusting groove **292h** may be spaced apart from the main fastening groove **2511h**. The main fastening groove **2511h** may be disposed between the outer weight adjusting groove **292h** and the inner weight adjusting groove **291h**. The outer weight adjusting groove **292h** may be connected to the other end of the auxiliary fastening groove **2512h** along the length direction. The auxiliary fastening groove **2512h** may be disposed between the outer weight adjusting groove **292h** and the inner weight adjusting groove **291h**. The outer end of the outer weight adjusting groove **292h** may be configured to be spaced apart from an outer seal ring insertion groove **254h**. Accordingly, the outer seal ring insertion groove **254h** may be disposed at a region between the outer weight adjusting groove **292h** and an upper portion of an outer surface of the retainer ring **23h**.

[0185] Because the remaining structure of the retainer ring **23h** is the same as or similar to that of the retainer ring **23** described above in FIGS. 2 to 7, that of the retainer ring **23a** described above in FIG. 8, or that of the retainer ring **23b** described above in FIG. 9, a repeated description or corresponding elements thereof is omitted.

[0186] In addition, because a structure in which an inner seal ring and an outer seal ring are disposed is the same as or similar to a disposition structure of the seal ring **271** and

272 described above in FIGS. 2 to 7, a disposition structure of the seal ring **271c** and **272c** described above in FIG. 10, a disposition structure of the seal ring **271d** and **272d** described above in FIG. 11, a disposition structure of the seal ring **271e** and **272e** described above in FIG. 12, a disposition structure of the seal ring **271f** and **272f** described above in FIG. 13, or a disposition structure of the seal ring **271g** and **272g** described above in FIG. 14, a repeated description of corresponding elements thereof is omitted.

[0187] FIG. 16 is an upper perspective view of a retainer ring **23i** according to another embodiment.

[0188] Referring to FIG. 16, a weight adjusting groove **252i** may be disposed at an upper surface of the retainer ring **23i**. A plurality of weight adjusting grooves **252i** may be configured to be spaced apart from each other along a circumferential direction. When viewed from above, an individual area of each weight adjusting groove **252i** may be larger than an individual area of each fastening groove **2511i** and **2512i**. The fastening groove **2511i** and **2512i** may include the main fastening groove **2511i** and the auxiliary fastening groove **2512i**. The weight adjusting groove **252i** may have a predetermined length along the circumferential direction. The weight adjusting groove **252i** may have an arc shape. Lengths of the weight adjusting grooves **252i** may correspond to each other. The weight adjusting groove **252i** may have a straight line shape in which an end portion of the weight adjusting groove **252i** along a length direction crosses the length direction. As an example, the end portion of the weight adjusting groove **252i** along the length direction may have the straight line shape facing a radial direction thereof. Additionally, one end of at least one of the plurality of weight adjusting grooves **252i** along the length direction may be rounded, and the other end of at least one of the plurality of weight adjusting grooves **252i** along the length direction may have a straight line shape in which the other end of at least one of the plurality of weight adjusting grooves **252i** along the length direction crosses the length direction.

[0189] As indicated above, according to example embodiments, the weight adjusting groove has an arc shape with a predetermined length along a circumferential direction. In further examples, the retainer ring may include a plurality of weight adjusting grooves, wherein each weight adjusting groove of the plurality of weight adjusting grooves has a length corresponding to each other weight adjusting groove. In other example embodiments, the weight adjusting groove is one of a plurality of weight adjusting grooves, in which at least one of the weight adjusting grooves has a length that is different from at least one other of the plurality of weight adjusting grooves.

[0190] Because the remaining structure of the weight adjusting groove **252i** is the same as or similar to that of the weight adjusting groove **252** described above in FIGS. 2 to 7, a repeated description of corresponding elements thereof is omitted.

[0191] Additionally, because the remaining structure of the retainer ring **23i** is the same as or similar to that of the retainer ring **23** described above in FIGS. 2 to 7, that of the retainer ring **23a** described above in FIG. 8, or that of the retainer ring **23b** described above in FIG. 9, a repeated description of corresponding elements thereof is omitted.

[0192] In addition, because a structure in which an inner seal ring and an outer seal ring are disposed is the same as or similar to a disposition structure of the seal ring **271** and

272 described above in FIGS. 2 to 7, a disposition structure of the seal ring 271c and 272c described above in FIG. 10, a disposition structure of the seal ring 271d and 272d described above in FIG. 11, a disposition structure of the seal ring 271e and 272e described above in FIG. 12, a disposition structure of the seal ring 271f and 272f described above in FIG. 13, or a disposition structure of the seal ring 271g and 272g described above in FIG. 14, a repeated description of corresponding elements thereof is omitted.

[0193] According to example embodiments, a seal ring is disposed between the retainer ring and the carrier body. The seal ring may include an inner seal ring disposed at a region between the weight adjusting groove and an upper end portion of an inner surface of the retainer ring; and an outer seal ring disposed at a region between the weight adjusting groove and an upper end portion of an outer surface of the retainer ring.

[0194] FIG. 17 is an upper perspective view of a retainer ring 23j according to another embodiment.

[0195] Referring to FIG. 17, a weight adjusting groove 252j may be disposed on an upper surface of the retainer ring 23j. A plurality of weight adjusting grooves 252j may be spaced apart from each other along a circumferential direction. When viewed from above, an individual area of each weight adjusting groove 252j may be larger than an individual area of each fastening groove 2511j and 2512j. The fastening groove 2511j and 2512j may include the main fastening groove 2511j and the auxiliary fastening groove 2512j. The weight adjusting groove 252j, such as a weight adjusting groove, may have a predetermined length along the circumferential direction. The weight adjusting groove 252j may have an arc shape. A length of at least one of the plurality of weight adjusting grooves 252j may be different from those of some of the remaining weight adjusting grooves 252j. The weight adjusting groove 252j may have a straight line shape in which an end portion of the weight adjusting groove 252j along a length direction crosses the length direction. As an example, the end portion of the weight adjusting groove 252j along the length direction may have the straight line shape facing a radial direction thereof. Additionally, an end portion of the weight adjusting groove 252j a length direction may be rounded. Additionally, an end portion of at least one of the plurality of weight adjusting grooves 252j along the length direction may be rounded, and end portions of the remaining weight adjusting grooves 252i along the length direction may have a straight line shape in which the end portions of the remaining weight adjusting grooves 252i along the length direction cross the length direction. Additionally, one end of at least one of the plurality of weight adjusting grooves 252j along the length direction may be rounded, and the other end of at least one of the plurality of weight adjusting grooves 252j along the length direction may have a straight line shape in which the other end of at least one of the plurality of weight adjusting grooves 252j along the length direction crosses the length direction.

[0196] Because the remaining structure of the weight adjusting groove 252j is the same as or similar to that of the weight adjusting groove 252 described above in FIGS. 2 to 7, a repeated description of corresponding elements thereof is omitted.

[0197] Additionally, because the remaining structure of the retainer ring 23j is the same as or similar to that of the retainer ring 23 described above in FIGS. 2 to 7, that of the

retainer ring 23a described above in FIG. 8, or that of the retainer ring 23b described above in FIG. 9, a repeated description of corresponding elements thereof is omitted.

[0198] In addition, because a structure in which an inner seal ring and an outer seal ring are disposed is the same as or similar to a disposition structure of the seal ring 271 and 272 described above in FIGS. 2 to 7, a disposition structure of the seal ring 271c and 272c described above in FIG. 10, a disposition structure of the seal ring 271d and 272d described above in FIG. 11, a disposition structure of the seal ring 271e and 272e described above in FIG. 12, a disposition structure of the seal ring 271f and 272f described above in FIG. 13, or a disposition structure of the seal ring 271g and 272g described above in FIG. 14, a repeated description of corresponding elements thereof is omitted.

[0199] FIG. 18 is an upper perspective view of a retainer ring 23k according to another embodiment.

[0200] Referring to FIG. 18, a weight adjusting groove 252k may be disposed at an upper surface of the retainer ring 23k. The weight adjusting groove 252k may include an inner weight adjusting groove 291k and an outer weight adjusting groove 292k.

[0201] The inner weight adjusting groove 291k may be disposed at an inner region of the upper surface of the retainer ring 23k along a radial direction. Because the remaining structure of the inner weight adjusting groove 291k is the same as or similar to that of the weight adjusting groove 252 described above in FIGS. 2 to 7, that of the weight adjusting groove 252i described above in FIG. 16, or that of the weight adjusting groove 252j described above in FIG. 17, a repeated description of corresponding elements thereof is omitted.

[0202] The outer weight adjusting groove 292k may be disposed at an outer region of the upper surface of the retainer ring 23k along a radial direction. The outer weight adjusting groove 292k may be disposed outside the radial direction with respect to the inner weight adjusting groove 291k. A length of the outer weight adjusting groove 292k may be the same as or different from a length of the inner weight adjusting groove 291k. A size of a central angle of the outer weight adjusting groove 292k may be the same as or different from a size of a central angle of the inner weight adjusting groove 291k. Because the remaining structure of the outer weight adjusting groove 292k is the same as or similar to that of the weight adjusting groove 252 described above in FIGS. 2 to 7, that of the weight adjusting groove 252i described above in FIG. 16, or that of the weight adjusting groove 252j described above in FIG. 17, a repeated description of corresponding elements thereof is omitted.

[0203] Additionally, because the remaining structure of the retainer ring 23k is the same as or similar to that of the retainer ring 23 described above in FIGS. 2 to 7, that of the retainer ring 23a described above in FIG. 8, or that of the retainer ring 23b described above in FIG. 9, a repeated description of corresponding elements thereof is omitted.

[0204] In addition, because a structure in which an inner seal ring and an outer seal ring are disposed is the same as or similar to a disposition structure of the seal ring 271 and 272 described above in FIGS. 2 to 7, a disposition structure of the seal ring 271c and 272c described above in FIG. 10, a disposition structure of the seal ring 271d and 272d described above in FIG. 11, a disposition structure of the seal ring 271e and 272e described above in FIG. 12, a disposition structure of the seal ring 271f and 272f described above in

FIG. 13, or a disposition structure of the seal ring 271g and 272g described above in FIG. 14, a repeated description of corresponding elements thereof is omitted.

[0205] FIG. 19 is a flow chart depicting an example manufacturing method in accordance with example embodiments.

[0206] Also provided herein are methods of manufacturing a semiconductor device. Example methods are provided herein which include use the substrate carrier heads and/or substrate polishing apparatuses provided herein.

[0207] FIG. 19 shows example methods provided herein. As shown in FIG. 19, example methods may include attaching a substrate to the substrate carrier head; depositing slurry onto the platen; performing etching on a surface of the substrate by moving the substrate carrier head to cause the substrate to contact the slurry and rotating one or more of the platen and the substrate carrier head; forming a layer on the substrate as a result of the polishing; and forming a semiconductor device on the substrate by adding more layers to form an integrated circuit on the substrate.

[0208] The substrate or substrate retainer may be attached to the substrate carrier head.

[0209] In the present methods, slurry is deposited onto the platen. In example embodiments, slurry may be applied onto a rotating polishing platen through a fluid control device, such as a metering pump or mass-flow-control regulator system. The slurry can be brought to the polishing platen for example, in a single-pass distribution system.

[0210] Etching may then be performed on a surface of the substrate by moving the substrate carrier head to cause the substrate to contact the slurry and rotating one or more of the platen and the substrate carrier head. The etching may selectively remove material from a thin film on a substrate (with or without prior structures on its surface) and by this removal create a pattern of that material on the substrate.

[0211] One or more layers may then be formed on the substrate as a result of polishing.

[0212] According to example methods, a semiconductor device is formed on the substrate by adding more layers to form an integrated circuit on the substrate.

[0213] While example embodiments are described herein, in connection with what is presently considered to be practical embodiments, it is to be understood that the disclosure is not limited to the disclosed embodiments, but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the spirit and scope of the appended claims.

What is claimed is:

1. A substrate carrier head comprising:
 - a carrier body; and
 - a retainer ring coupled to the carrier body, wherein the retainer ring comprises a weight adjusting groove recessed into the retainer ring by a predetermined depth, wherein the retainer ring has an upper surface facing the carrier body, and wherein the weight adjusting groove is disposed at the upper surface of the retainer ring.
2. The substrate carrier head of claim 1, wherein the weight adjusting groove is one of a plurality of weight adjusting grooves, wherein the plurality of weight adjusting grooves are spaced apart from each other along a circumferential direction of the retainer ring.

3. The substrate carrier head of claim 2, wherein a fastening groove recessed downward into the retainer ring by a predetermined depth is disposed at the upper surface of the retainer ring, and wherein the carrier head further includes a fastening member that protrudes downward from the carrier body toward the upper surface of the retainer ring, the fastening member being configured to be inserted into the fastening groove.
4. The substrate carrier head of claim 3, wherein the fastening groove is one of a plurality of fastening grooves, wherein each of the fastening grooves is spaced apart from each other of the fastening grooves along the circumferential direction of the carrier head.
5. The substrate carrier head of claim 4, wherein at least one of the plurality of weight adjusting grooves is disposed between adjacent fastening grooves of the plurality of fastening grooves.
6. The substrate carrier head of claim 2, wherein the plurality of weight adjusting grooves comprises:
 - an inner weight adjusting groove disposed at an inner region of the upper surface of the retainer ring along a radial direction; and
 - an outer weight adjusting groove disposed at an outer region of the upper surface of the retainer ring along the radial direction.
7. The substrate carrier head of claim 1, wherein the weight adjusting groove has an arc shape with a predetermined length along a circumferential direction.
8. The substrate carrier head of claim 7, wherein the weight adjusting groove is one of a plurality of weight adjusting grooves, wherein each weight adjusting groove of the plurality of weight adjusting grooves has a length corresponding to each other weight adjusting groove.
9. The substrate carrier head of claim 7, wherein the weight adjusting groove is one of a plurality of weight adjusting grooves, wherein at least one of the weight adjusting grooves has a length that is different from at least one other of the plurality of weight adjusting grooves.
10. The substrate carrier head of claim 1, wherein the weight adjusting groove has a ring shape.
11. The substrate carrier head of claim 10, wherein the weight adjusting groove is one of a plurality of weight adjusting grooves comprising:
 - an inner weight adjusting groove disposed at an inner region of the upper surface of the retainer ring along a radial direction; and
 - an outer weight adjusting groove disposed at an outer region of the upper surface of the retainer ring along the radial direction.
12. The substrate carrier head of claim 1, further comprising a seal ring disposed between the retainer ring and the carrier body.
13. The substrate carrier head of claim 12, wherein the seal ring comprises:
 - an inner seal ring disposed at a region between the weight adjusting groove and an upper end portion of an inner surface of the retainer ring; and
 - an outer seal ring disposed at a region between the weight adjusting groove and an upper end portion of an outer surface of the retainer ring.
14. The substrate carrier head of claim 1, further comprising a supplementary member disposed inside the weight adjusting groove.

15. The substrate carrier head of claim **1**, wherein the retainer ring comprises an upper ring disposed at an upper region of the retainer ring and a lower ring disposed at a lower region of the retainer ring, wherein the weight adjusting groove is disposed only at the upper ring.

16. A substrate carrier head comprising:

a carrier body;

a retainer ring coupled to the carrier body;

a fastening member that protrudes from the carrier body toward the retainer ring, which fastening member is configured to fit into a fastening groove of the retainer ring; and

a seal ring disposed between the retainer ring and the carrier body,

wherein the retainer ring has an upper surface facing the carrier body, and

wherein the fastening groove recessed into the retainer ring by a predetermined depth and a weight adjusting groove recessed into the retainer ring by a predetermined depth are disposed at the upper surface of the retainer ring.

17. The substrate carrier head of claim **16**, wherein the weight adjusting groove is one of a plurality of weight adjusting grooves, wherein the plurality of weight adjusting grooves are spaced apart from each other along a circumferential direction of the retainer ring.

18. The substrate carrier head of claim **16**, wherein the weight adjusting groove has a ring shape.

19. The substrate carrier head of claim **18**, wherein the weight adjusting groove is one of a plurality of weight adjusting grooves, wherein the plurality of weight adjusting grooves comprising

an inner weight adjusting groove disposed at an inner region of the upper surface of the retainer ring along a radial direction; and

an outer weight adjusting groove disposed at an outer region of the upper surface of the retainer ring along the radial direction,

wherein the fastening groove is disposed between the inner weight adjusting groove and the outer weight adjusting groove.

20. A substrate polishing apparatus, comprising:

a platen; and

a substrate carrier head disposed adjacent to the platen, wherein the substrate carrier head comprises

a carrier body,

a retainer ring coupled to the carrier body,

a fastening member that protrudes from the carrier body toward the retainer ring, which fastening member is configured to fit into a fastening groove of the retainer ring,

an inner seal ring disposed between the retainer ring and the carrier body at a region between a weight adjusting groove and an upper end portion of an inner surface of the retainer ring, and

an outer seal ring disposed between the retainer ring and the carrier body at a region between the weight adjusting groove and an upper end portion of an outer surface of the retainer ring,

wherein the retainer ring has an upper surface facing the carrier body, and

wherein the fastening groove recessed into the retainer ring by a predetermined depth and the weight adjusting groove recessed into the retainer ring by a predetermined depth, are disposed at the upper surface of the retainer ring, and

wherein the fastening member is inserted into the fastening groove.

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